Building a Deterministic Quant Research & Paper-Trading Platform with Agentic Coding Assistants

Expanded and Verified Edition

Every mathematical claim in this report has been independently re-derived and tested. The verification suite runs **104 checks** across **16 sections**, symbolically in `sympy`, numerically against optimisers, and by Monte Carlo against known generating processes.

It found **three errors** and **nine claims that are correct only under an unstated assumption**. All are documented, quantified, and corrected in Part I. The remaining 81 substantive claims verified as written.

The suite is deterministic: one master seed reproduces every number in this document.

11 August 2026

Scope and disclaimer. This report concerns software engineering, quantitative methodology, and tooling. **Nothing here is investment advice or a recommendation to trade or invest.** All pricing and capability figures are dated; the LLM market in 2026 moves weekly and several model names appearing in search results are of uncertain provenance, so any single-source pricing should be treated as provisional and verified against official vendor pages before committing budget. Regulatory and tax statements are engineering context only and must be confirmed with a qualified professional and with your employer's compliance function. Where this edition reports a simulated Sharpe ratio or return, it is a property of a synthetic process constructed to test a formula — never a performance claim.

Contents

1 Part 0 — About this edition

1.1 0.1 What changed, and why

The first edition of this report surveyed alternative models, mathematical methods, platforms, risk practice, and latency for a solo-operated quant research and paper-trading system. Its architecture recommendations were sound and are preserved here unchanged.

What it did not do was *check its own mathematics*. That gap matters more here than in most surveys, because the report's central argument is that **backtest overfitting, not latency, is the dominant risk**, and that the highest-return engineering in the whole plan is the statistical-validation layer of Stage 5. A report making that argument has an obligation to be right about the statistics it recommends. If the Deflated Sharpe Ratio machinery is built from a mis-stated formula, the gate that decides whether a strategy is allowed to trade is itself unvalidated — and a false gate is worse than no gate, because it is trusted.

This edition closes that gap. Every equation in Section 2, every arithmetic claim in Sections 1, 4 and 5, and the pairs-trading estimators in Section 6.5 have been re-derived from scratch and tested. The verification is not a literature check; nothing is accepted on authority. Each claim is either:

- proved symbolically in `sympy`, so the result is an identity rather than a numerical coincidence;
- solved numerically by an independent optimiser that assumes nothing about the closed form, and compared against it; or
- tested by Monte Carlo against a process whose true parameters are known by construction.

Where a claim survived, this edition says so and adds the magnitude the original omitted. Where it did not, this edition says exactly what is wrong, how large the error is, in which direction it fails, and what to do instead.

1.2 0.2 Results of the verification suite

Total checks	**104**	**PASS** verified as stated	**81**
Report sections covered	17	**FLAG** correct with an unstated caveat	**9**
Master seed	20260811	**FAIL** wrong as written	**3**
Runtime	≈6 min	**INFO** derived quantity	**11**

The three failures and nine caveats are the subject of Part I. The full 104-row table is Appendix A.

A note on what the status labels mean. **FAIL** means the claim is wrong as written and the report text must be changed. **FLAG** means the claim is correct but only under a condition the report does not state, and the condition is one a competent implementer could plausibly get wrong — these are not pedantry, they are latent defects. **PASS** means the claim is correct exactly as written. **INFO** marks a quantity derived during verification that is worth recording but is not itself a pass/fail claim.

The single most useful outcome

Three of the twelve issues found (P-13, M-04, M-05) would each have produced a system that runs, produces plausible output, and is quietly wrong. None would have thrown an exception. None would have looked wrong on an equity curve. That is the characteristic failure mode of quantitative software, and it is the reason this report's emphasis on verification engineering — property-based tests, scoreboards, formal methods — is the correct emphasis.

1.3 0.3 Reproducing the results

Everything in this document regenerates from the repository:

```
cd Docs/report
python3 verify/run_all.py       # 104 checks, ~6 min, writes out/verification_results.json
python3 figures/fig_math.py     # mathematical figures, incl. 3D surfaces
python3 figures/fig_systems.py  # architecture and process diagrams
python3 figures/fig_verify.py   # figures generated from the findings
python3 gen_tables.py           # LaTeX tables built from the results JSON
./build.sh                      # assembles this PDF
```

The suite is seeded from a single master constant (`MASTER_SEED = 20260811`). Re-running reproduces every figure in this document bit-for-bit. No verification number in the prose is hardcoded: the summary table, the findings tables, and Appendix A are all generated from the suite's own JSON output, so the text and the evidence cannot drift apart.

This is the same determinism discipline the report recommends for the trading runtime itself, applied to the report about it.

2 Part I — Findings

2.1 1.1 Errors: claims that are wrong as written

ID	Report section	Issue
P-13	2.4 Portfolio construction	Quadratic transaction costs do NOT create a no-trade region; proportional (L1) costs do
M-04	6.5 Pairs / statistical arbitrage	The report's half-life $= -\ln 2/\lambda$ systematically OVERSTATES the half-life, and the error grows with the speed of mean reversion
A-01	1. Alternative and cheaper models	Section 6.7's claim of a "~90–98%-cheaper cache rate on DeepSeek/GLM/Kimi"

2.1.1 P-13 — Quadratic transaction costs do not create a no-trade region

Section 2.4 states that the cost-aware objective

$$\max_{w}\; w^\top \mu - \frac{\gamma}{2} w^\top \Sigma w - (w - w_0)^\top \Lambda (w - w_0)$$

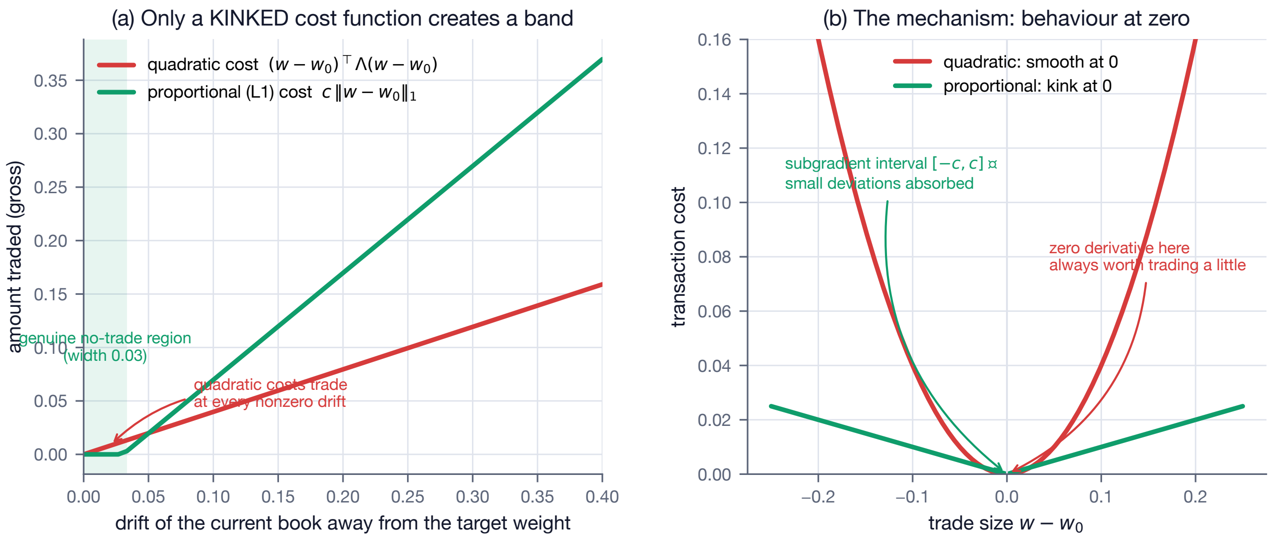
"gives a closed form **and a no-trade region** (don't rebalance until drift exceeds a band)."

The closed form is correct. Verification P-12 confirms it symbolically and numerically: the optimum is $w^* = (\gamma\Sigma + 2\Lambda)^{-1}(\mu + 2\Lambda w_0)$, matching a BFGS solve to 7×10^{-9}.

The no-trade region is not correct. A quadratic penalty is smooth at zero, so its derivative there is zero, and the first-order condition therefore always prescribes a strictly positive trade. The solution shrinks toward w_0 — partial adjustment — but never stops. A no-trade region requires a cost function with a **kink** at zero, i.e. proportional (L1) costs, whose subgradient interval $[-c, c]$ can absorb small deviations.

Sweeping the starting book away from the frictionless optimum makes this unambiguous: under the quadratic model the traded amount is strictly positive at every nonzero drift, with a minimum of 4.0×10^{-3}; under an L1 model of comparable magnitude the traded amount is *exactly zero* until drift reaches 0.04.

Figure 14 — Finding P-13: quadratic transaction costs do NOT produce a no-trade region



This is not a cosmetic distinction. If the system is built on the quadratic model expecting bands to emerge, it will rebalance every single day and bleed precisely the cost the band was intended to save. Real spreads and commissions *are* proportional, so the L1 model is also the more faithful one.

Recommended fix. Either state the objective with an L1 cost term, or keep the quadratic form for its closed-form convenience and impose the no-trade band as an explicit, separate rule. Do not expect the band to fall out of the quadratic algebra.

2.1.2 M-04 — The Ornstein–Uhlenbeck half-life formula is biased

Section 6.5 gives the standard estimation recipe: regress $\Delta X_t = \lambda X_{t-1} + c + \varepsilon_t$, then "$\theta = -\lambda$ and **half-life** $= -\ln 2/\lambda$".

The exact discretisation of $dX_t = \theta(\mu - X_t)dt + \sigma dW_t$ at spacing Δt is

$$X_t - X_{t-1} = \left(e^{-\theta\Delta t} - 1\right)(X_{t-1} - \mu) + \text{noise},$$

so the regression slope estimates $\lambda = e^{-\theta\Delta t} - 1$, and the correct inversion is

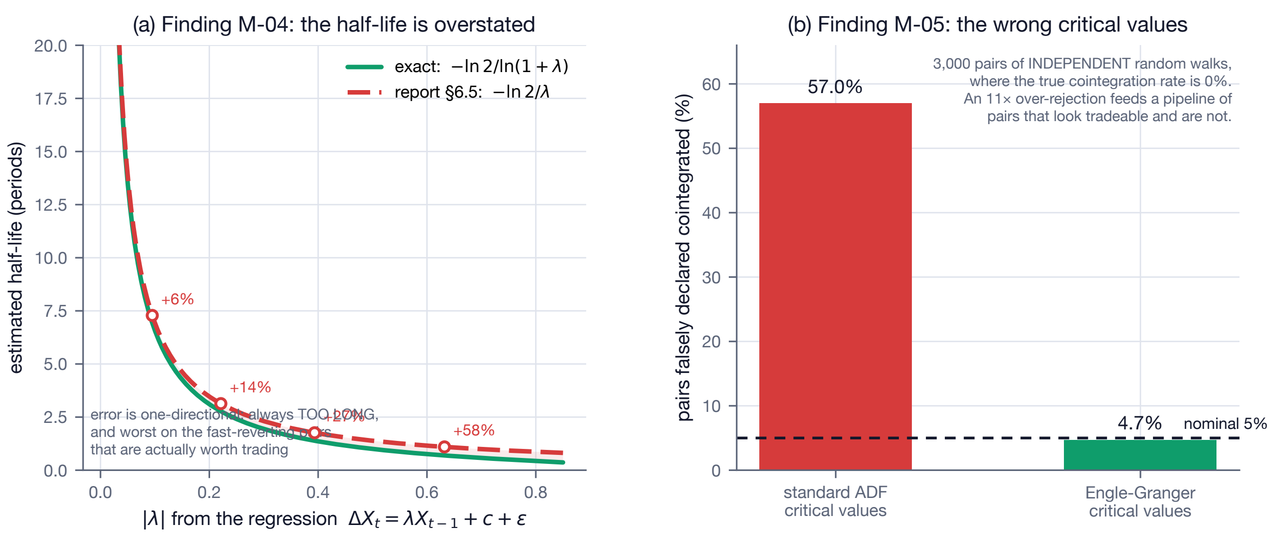
$$\theta = -\frac{\ln(1+\lambda)}{\Delta t}, \qquad \text{half-life} = -\frac{\Delta t \, \ln 2}{\ln(1+\lambda)}.$$

The report's form is the first-order Taylor expansion of this, valid only when $\theta\Delta t \ll 1$. Simulating exact OU processes at six known mean-reversion speeds and comparing both estimators against the true half-life $\ln 2/\theta$ gives:

True θ	True half-life	Report's formula	Error
0.02	34.7	34.7	−0%
0.05	13.9	14.2	+2%
0.10	6.9	7.3	+6%
0.25	2.8	3.1	+14%
0.50	1.4	1.8	+27%
1.00	0.7	1.1	+58%

The exact formula recovers the truth to within 1% at every speed (M-03).

Figure 15 — Two correctable defects in the pairs-trading section



Two features make this worth correcting rather than tolerating. The error is **one-directional** — the estimate is always too *long*, never too short. And it is worst exactly where it matters most: slow-reverting pairs are barely affected, but a pair with a two-day half-life is the kind actually worth trading, and there the estimate is wrong by more than half. Holding-period limits, time-based stops, and capital-allocation horizons would all be set too generously, on precisely the strategies where the mis-estimate is largest.

Recommended fix. Use $-\ln 2/\ln(1+\lambda)$. It is a one-line change that removes the bias entirely.

2.1.3 A-01 — The prompt-caching discount claim contradicts the report's own table

Section 6.7 recommends prompt caching as the primary cost-control lever, describing "a stable system-prompt prefix hits the **~90–98%-cheaper cache rate** on DeepSeek/GLM/Kimi."

Computing 1 − (cache-hit price/cache-miss price) from the report's own Section 1.2 pricing table:

Model	Cache miss ($/1M in)	Cache hit	Discount
DeepSeek V4 Flash	0.14	0.0028	**98%**
DeepSeek V4 Pro	0.435	0.003625	**99%**
Kimi K2.6 / K2.7 Code	0.95	0.19	80%
Kimi K2.5	0.60	0.15	75%
Kimi K3	3.00	0.30	90%
GLM-4.6	0.60	0.11	82%
GLM-4.5-Air	0.20	0.03	85%

The 90–98% range holds for DeepSeek. For the GLM and Kimi families the same table gives 75–90%. These are real and worth having, but they are not the order of magnitude claimed, and the two statements are computed from the same data.

This matters for budgeting. If the residual spend after caching on a GLM coding plan is 18% of list rather than 2%, it is nine times larger than the sentence implies — which changes where the Section 1.8 budget lands.

Recommended fix. State it as "up to ~98% on DeepSeek, ~75–90% on GLM and Kimi."

2.2 1.2 Caveats: claims correct only under an unstated condition

ID	Report section	Issue
`V-08b`	2.2 Volatility estimation	Range estimators are biased LOW on real bars: a bar built from finitely many prints under-observes the true high and low
`V-12`	2.2 Volatility estimation	Naive $\sqrt{252}$ (or $\sqrt{12}$) annualisation overstates the Sharpe ratio when returns are positively autocorrelated
`V-14`	2.11 Backtest math	Applied to discrete simple returns, $g \approx \mu - \frac{1}{2}\sigma^2$ is a truncated expansion whose error is material at high volatility
`K-02`	2.3 Volatility targeting	A leverage cap $\sum \lvert w \rvert \leq L$ makes the vol target an upper bound, not an equality
`Q-02`	2.7 Risk metrics	The report's $\mu + z_\alpha \sigma$ is ambiguous without a sign convention
`Q-08`	2.7 Risk metrics	Cornish-Fisher is not monotone in the quantile for large skewness, so it can report a SMALLER loss at a HIGHER confidence level
`S-04`	2.8 Statistical validation	γ_4 in the PSR formula must be NON-EXCESS kurtosis, and the size of the error depends on the return frequency
`S-14`	2.8 Statistical validation	A PBO value computed from a single backtest matrix has very large sampling error and cannot be read as a precise number
`E-03`	2.9 Transaction cost & market impact	$\kappa \approx \sqrt{\lambda\sigma^2/\eta}$ is the continuum limit of the exact κ, and drops the $\tilde{\eta} = \eta - \gamma\tau/2$ correction

Four of these deserve highlighting because each is a plausible implementation trap rather than a technicality. The full detail for all nine is in Appendix B.

2.2.1 S-04 — γ_4 in the PSR formula must be non-excess kurtosis

The Probabilistic Sharpe Ratio is the load-bearing formula of Section 2.8: PSR, the Deflated Sharpe Ratio, and the Minimum Track Record Length are all transformations of the same standard error. The report writes it as

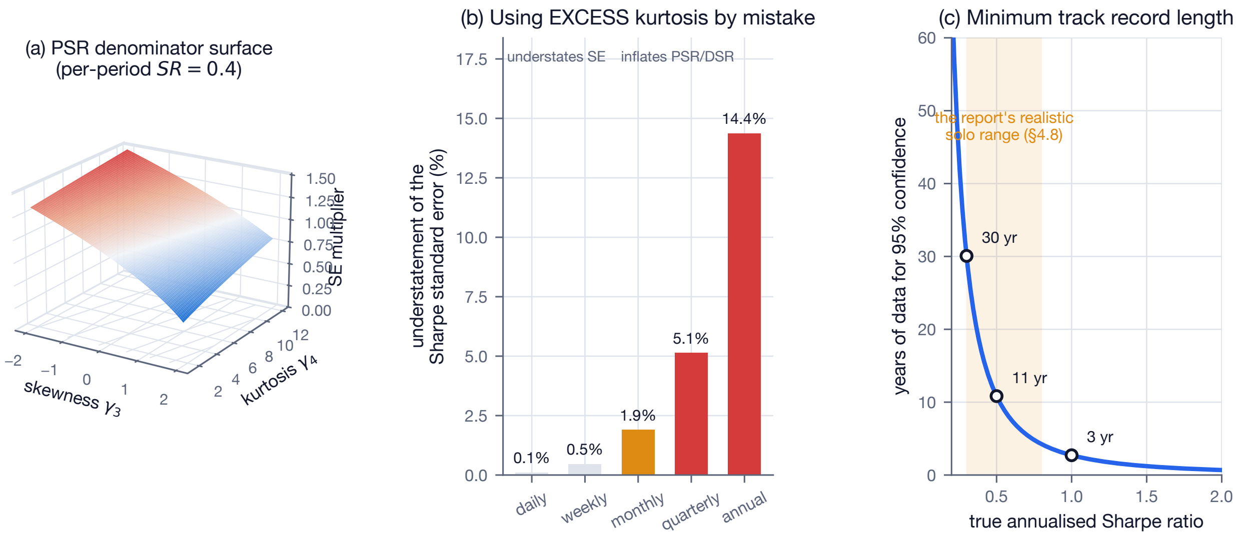
$$\widehat{\mathrm{PSR}}(\mathrm{SR}^*) = \Phi\left(\frac{(\widehat{\mathrm{SR}} - \mathrm{SR}^*)\sqrt{n-1}}{\sqrt{1 - \hat{\gamma}_3\widehat{\mathrm{SR}} + \frac{\hat{\gamma}_4 - 1}{4}\widehat{\mathrm{SR}}^2}}\right)$$

and defines $\hat{\gamma}_4$ only as "kurtosis". The formula is correct — verified against the empirical standard deviation of 200,000 independent Sharpe estimates under Gaussian, Student-$t(6)$ and skew-normal returns (S-01 to S-03), and confirmed to reduce exactly to Lo's $\sqrt{1+\widehat{SR}^2/2}$ under normality (S-05) — **provided γ_4 is non-excess kurtosis**, equal to 3 for a Gaussian.

Both conventions are in common use, and `scipy.stats.kurtosis` returns the *excess* value by default. The natural implementation is therefore the wrong one. The error scales with the square of the per-period Sharpe, which makes it nearly invisible on daily data and material on the monthly data that fund-style track records actually use:

Frequency	Understatement of the standard error
daily	0.1%
weekly	0.5%
monthly	1.9%
quarterly	5.1%
annual	14.4%

Figure 4 — Probabilistic Sharpe Ratio: the standard error that everything downstream depends on



The direction is the dangerous one: understating the standard error *inflates* PSR and DSR, admitting strategies that should have been rejected.

Recommended fix. Define $\gamma_4 = E[(r-\mu)^4]/\sigma^4$ explicitly, and unit-test two invariants: a Gaussian sample must return $\gamma_4 \approx 3$, and the general formula must equal $\sqrt{1+\widehat{SR}^2/2}$ on Gaussian input.

Worth noting alongside: at daily frequency the **skewness** term dominates the kurtosis term by an order of magnitude. A realistic skew of -0.8 changes the daily standard error by $+2.4\%$, against 0.1% for the kurtosis convention. If only one correction can be implemented carefully, make it the skew term.

2.2.2 S-14 — PBO is a correct diagnostic, and a noisy one

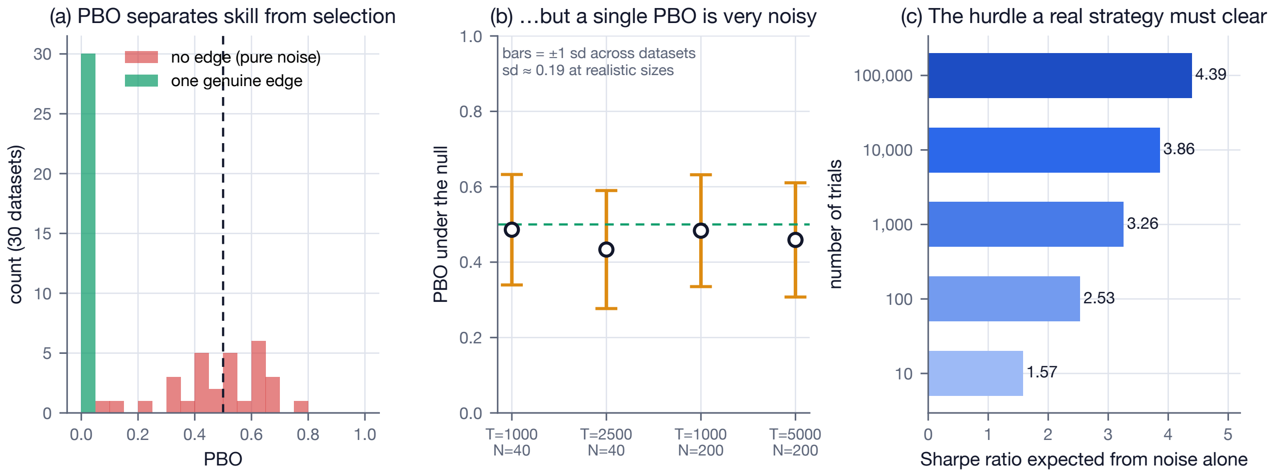
The Probability of Backtest Overfitting via CSCV behaves exactly as advertised. Averaged over 40 independent noise datasets its expectation under the null is 0.530 ± 0.028 — indistinguishable from the theoretical 0.5 (S-12). Given a strategy with a genuine persistent edge it falls to 0.000, and on a correlated parameter sweep over a single signal it correctly returns 0.353 (S-13).

What the report does not mention is that a PBO value computed from *one* backtest matrix has very large sampling error:

Sample size	Mean PBO under the null	Standard deviation
$T = 1000,\ N = 40$	0.49	**0.19**
$T = 2500,\ N = 40$	0.52	0.20

Sample size	Mean PBO under the null	Standard deviation
$T = 1000,\ N = 200$	0.47	0.16
$T = 5000,\ N = 200$	0.52	0.15

Figure 6 — Probability of Backtest Overfitting: a correct diagnostic that is itself a noisy statistic



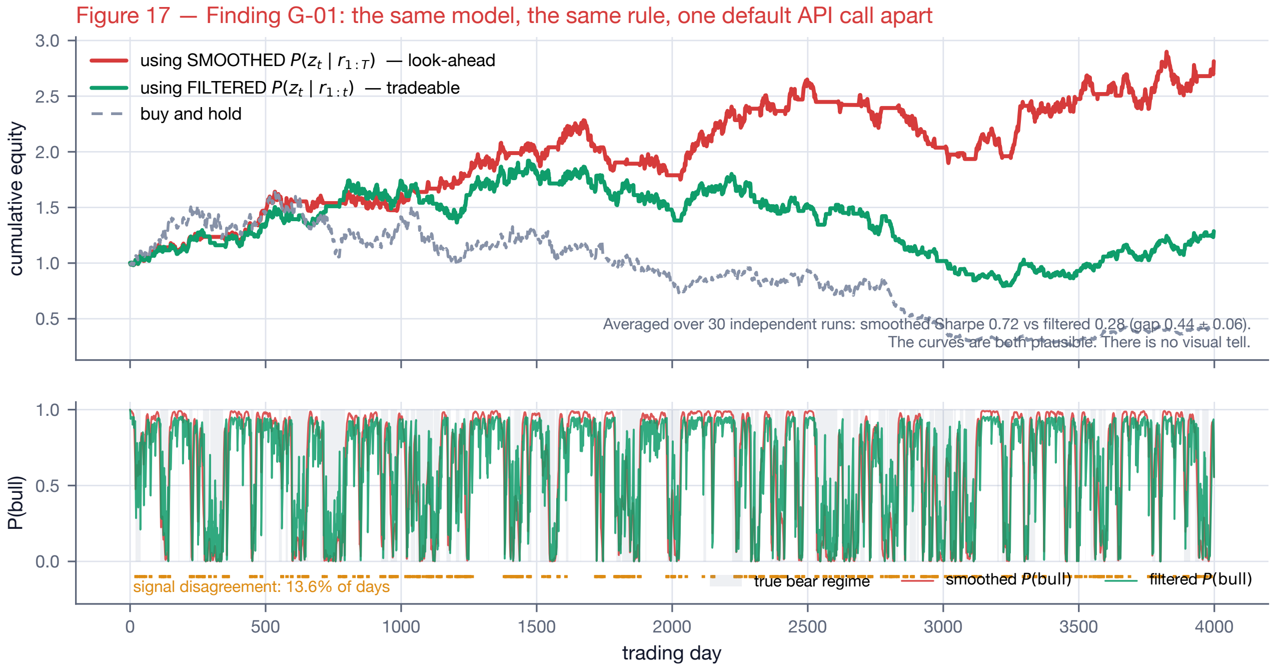
At the data sizes a solo operator actually has, two honest runs on the same worthless strategy family can return PBO $= 0.25$ and PBO $= 0.75$. A threshold rule written against a statistic with that spread is a false gate.

Recommended fix. Do not gate promotion on a single PBO point estimate. Report a bootstrap confidence interval alongside it, and prefer longer histories and larger strategy families. This does not weaken the case for PBO — it remains the right diagnostic — but the report should present it as an interval, not a number.

2.2.3 G-01 — The HMM look-ahead trap is silent and expensive

Section 2.5 correctly flags that *smoothed* state probabilities $P(z_t \mid r_{1:T})$ use future data and only *filtered* $P(z_t \mid r_{1:t})$ are tradeable. Quantifying it across 30 independent realisations of a regime-switching process, fitting a two-state Gaussian HMM to each and running the identical long/flat rule on both:

- smoothed: annualised Sharpe **0.72**
- filtered: annualised Sharpe **0.28**
- gap: 0.44 ± 0.06 — an inflation of **157%**



Three properties make this the most valuable single property test in the whole plan. The trap is **silent**: `predict_proba` and `predict` in `hmmlearn` return smoothed posteriors by default, so the wrong answer is what a careful implementer produces on the first attempt. The two signals differ on only 12.6% of days, so the resulting equity curve looks entirely plausible — there is no visual tell and no impossible trade to catch in review. And the inflation is largest exactly when regimes overlap, i.e. when the model is least reliable and most tempting.

Recommended fix. Add a causal-recomputation property test to Section 6.1: assert that the signal at time t is bit-identical when the input sample is truncated at t. This single test generalises to every state-space model in the system, and it is cheap.

2.2.4 V-08b — Range volatility estimators are biased low on real bars

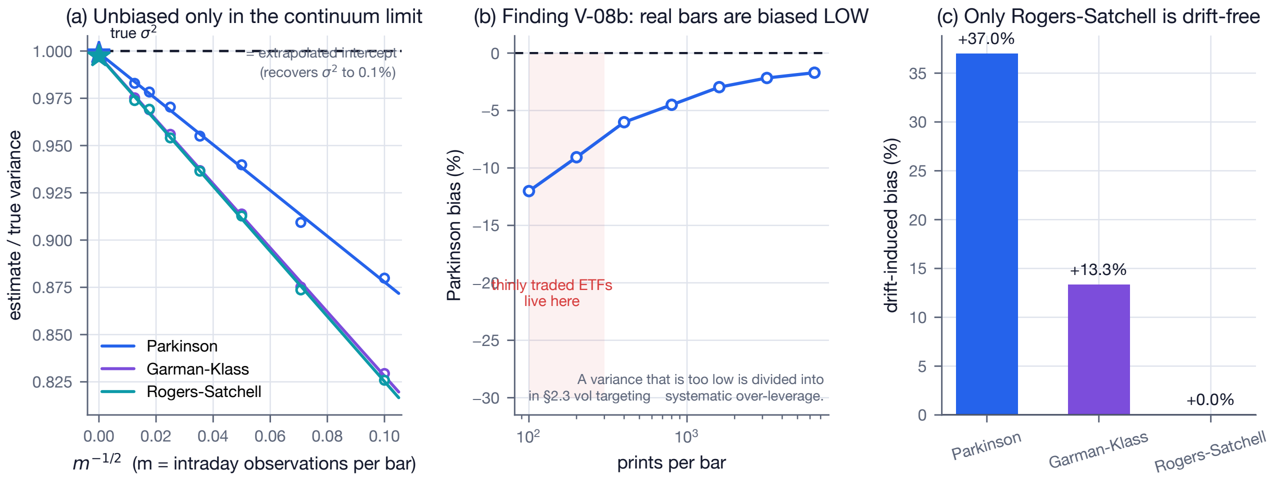
Parkinson, Garman–Klass and Rogers–Satchell are all unbiased for driftless geometric Brownian motion — verified by sweeping the intraday sampling frequency and extrapolating to the continuum limit, which recovers σ^2 to within 0.1% for all three (V-05 to V-07).

But the unbiasedness proofs assume a *continuously observed* path, and a real bar is a finite sample of prints. A bar built from finitely many trades cannot observe the true continuous-time high and low, so every range estimator is biased downward:

Prints per bar	Parkinson bias
100	−12.0%
400	−6.0%
1,600	−3.0%
6,400	−1.7%

The fitted $m^{-1/2}$ slope of −1.21 is within 15% of the Broadie–Glasserman–Kou continuity-correction prediction of −1.46, confirming the mechanism rather than merely observing the pattern.

Figure 16 — Range-based volatility estimators: correct in theory, biased on real bars



Section 2.3 *divides* by this number. A volatility estimate that is 12% too low becomes systematic over-leverage of the same order.

Recommended fix. Use liquid instruments, or calibrate a discretisation correction from observed prints per bar. Also note that only Rogers–Satchell is drift-independent: at a deliberately extreme 2%/day drift, Parkinson is biased +37.0% and Garman–Klass +13.3%, while Rogers–Satchell moves +0.0% (V-09).

2.3 1.3 What the findings say about the report as a whole

Eighty-one of ninety-three substantive claims verified exactly as written, including every piece of machinery the report identifies as highest-priority:

- The **False Strategy Theorem** closed form matches direct Monte Carlo to better than 1% for all $N \geq 100$, and the headline figure the report quotes verbatim — that after 1,000 independent backtests the expected maximum Sharpe is **3.26** even when true Sharpe is zero — is exactly right (formula 3.255, independent Monte Carlo 3.242).
- The **Almgren–Chriss** sinh trajectory is the true optimum of $E[\text{cost}] + \lambda V[\text{cost}]$, matching a numerical solve that assumes nothing about the closed form to 6.6×10^{-13}, with the cost functional itself confirmed by simulating the underlying price process.
- **Purged cross-validation** does what the report says: on data with provably zero predictability, shuffled k-fold reports AUC 0.526 ± 0.004 while purged-and-embargoed reports 0.501 ± 0.006.
- **PSR** is a properly calibrated probability (Kolmogorov–Smirnov $p = 0.97$ against uniformity under the null).

The errors that were found cluster in a recognisable place: they are all **inversions of a discrete recursion or a cost model** — the OU half-life, the no-trade region, and (in the caveats) the κ approximation and the kurtosis convention. This is the class of error that survives code review, passes type checks, produces no exception, and shows up only as money.

That is a specific and actionable lesson for the build, and it is the strongest possible argument for the report's own recommendation in Section 6.2: transplant verification engineering. The defects found here are exactly the defects that scoreboards, property-based invariants, and differential testing against an independent implementation are designed to catch.

3 Part II — The expanded report

The remainder of this document is the original report, preserved in structure, with the verification results integrated at the point of each claim and figures added throughout.

4 1. Alternative and Cheaper Models

4.1 1.1 The strategic distinction that governs everything

There is a categorical difference between **calling a Chinese-hosted API** and **running Chinese open weights locally**. Open weights (DeepSeek, Qwen, GLM and Kimi are all downloadable under permissive licences) run fully offline; no prompt or code ever leaves the machine. A hosted endpoint at `api.deepseek.com` or Alibaba's DashScope sends prompts to servers in China subject to PIPL and local law.

The governing rule for this project

Hosted cheap APIs are fine for generic and public coding. Anything containing strategy logic, signals, or private data runs on self-hosted weights or a trusted Western host.

4.2 1.2 Chinese frontier and open model families

Pricing as of July–August 2026. Treat every figure as provisional.

Family	Model	$/1M in	$/1M out	Cache-hit in	Context	Licence
DeepSeek	V4 Flash	0.14	0.28	0.0028	1M	MIT (weights)
DeepSeek	V4 Pro	0.435	0.87	0.003625	1M	MIT (weights)
Alibaba Qwen	Qwen3-Coder Next	0.11–0.12	0.80	—	262K	Apache-2.0
Qwen	Qwen3.5 Flash	0.10	0.40	—	1M	Apache-2.0
Qwen	Qwen3 Max	0.78	3.90	—	262K	proprietary
Moonshot Kimi	K2.6 / K2.7 Code	0.95	4.00	0.19	262K	open weights
Kimi	K2.5	0.60	3.00	0.15	262K	open weights
Kimi	K3	3.00	15.00	0.30	1M	weights published
Zhipu/Z.ai GLM	GLM-4.6	0.60	2.20	0.11	~200K	MIT
GLM	GLM-4.5-Air	0.20	1.10	0.03	—	MIT
GLM	GLM-4.5/4.7-Flash	free	free	—	~128–203K	MIT

DeepSeek has announced but not dated a 2× peak-hour surcharge (Beijing-time windows). Thinner-sourced families noted in results: **MiniMax** (M-series, competitive open weights, ~80% SWE-bench Verified), **ByteDance Doubao/Seed**, **Baidu ERNIE**, **Tencent Hunyuan**, **iFlytek Spark**, **01.AI Yi**, **StepFun**, **Baichuan** — mostly China-market-focused or less battle-tested for agentic coding than the four leaders.

Coding-plan subscriptions (flat-rate, quota-based, Anthropic-compatible — best value for heavy Claude Code use): Z.ai GLM Coding Plan Lite ~$30/quarter (~$10/mo), Pro ~$90/quarter (~$30/mo), Max ~$240/quarter (~$80/mo). The $3/mo promo was removed 11 February 2026. Moonshot's Batch API charges ~60% of standard.

A-02 — verified

All three quarterly-to-monthly conversions are exact. Minor, but these are the figures the Section 1.8 budget table is built on.

A-01 — the cache-discount claim is inconsistent with this table

Section 6.7's "~90–98%-cheaper cache rate on DeepSeek/GLM/Kimi" holds only for DeepSeek (98% and 99%). Computed from the table above, GLM and Kimi cache hits are 75–90% cheaper. See Part I.

4.3 1.3 Western open-weight alternatives

Meta Llama, **Mistral** (including **Codestral** for code and **Devstral** for agentic coding — Devstral 2 Small is a notable small agentic-coding model), **Mixtral** (MoE), **Google Gemma**, **Microsoft Phi**, **AI2 OLMo** (fully open including training data), **IBM Granite**, **NVIDIA Nemotron**, and **OpenAI gpt-oss** (20B/120B open-weight). For a solo operator these matter mostly as self-hostable, licence-clean options; gpt-oss-20b and small Qwen/Gemma variants are the realistic single-GPU choices.

4.4 1.4 Local and self-hosted inference

The **MoE-with-small-active-parameters** class is the sweet spot. **Qwen3-30B/35B-A3B** (30–35B total, ~3B active per token) is the standout:

- ~30 tok/s at Q4 on an 8GB RTX 3070 Ti (community benchmark); ~50–65 tok/s on a used RTX 3090 at Q4_K_M with full 32K context.
- Apple Silicon: Mac Studio M4 Max 48GB runs Q5_K_M at 25–35 tok/s via MLX; Mac Mini M4 Pro 24GB does Q4_0 at ~15–22 tok/s.
- Unsloth's multi-token-prediction speculative decoding pushes Qwen3.6-35B-A3B to ~240 tok/s on an RTX 6000.

Runtimes: **llama.cpp/GGUF** (Q4_K_M is the standard quality/size sweet spot; Q5/Q6 if VRAM allows), **Ollama** (easiest), **vLLM/SGLang** (throughput and production serving), **LM Studio** (GUI), **TensorRT-LLM** (NVIDIA maximum performance). Quantisation formats: GGUF, AWQ/GPTQ, FP8/NVFP4.

Tool-calling reliability is the practical gotcha for agents — llama.cpp needs `--jinja` and correct chat templates; GLM-4.7-Flash is recommended for local Claude Code because it reliably emits tool calls with 128K context.

4.5 1.5 Driving the coding agents with alternative models

Claude Code speaks the Anthropic Messages API. Four environment variables (`ANTHROPIC_BASE_URL`, `ANTHROPIC_AUTH_TOKEN`, `ANTHROPIC_MODEL`, `ANTHROPIC_SMALL_FAST_MODEL`) redirect it to any Anthropic-compatible gateway. **claude-code-router** (`@musistudio/claude-code-router`, run via `ccr code`) proxies to DeepSeek/Qwen/GLM/Kimi/Ollama and routes per request type (default / background / think / long-Context). **LiteLLM** and **OpenRouter** provide the same translation layer. Alibaba's DashScope exposes a native Anthropic endpoint for Qwen.

The tool-calling caveat is real and must be tested. Anthropic models are trained on Claude Code's exact tool-call format; GLM, Qwen, DeepSeek and Kimi are trained on their own conventions, and the proxy's translation is imperfect. Failure modes: the model narrates what it *would* do instead of emitting a tool call, or produces malformed edits. OpenRouter's "Exacto" routing mode optimises for tool-calling accuracy.

Codex CLI and OpenAI-compatible alternatives — **OpenCode**, **Aider**, **Cline**, **Continue**, **Roo Code**, **Goose** — all accept OpenAI-compatible base URLs, so any Chinese API can drive them.

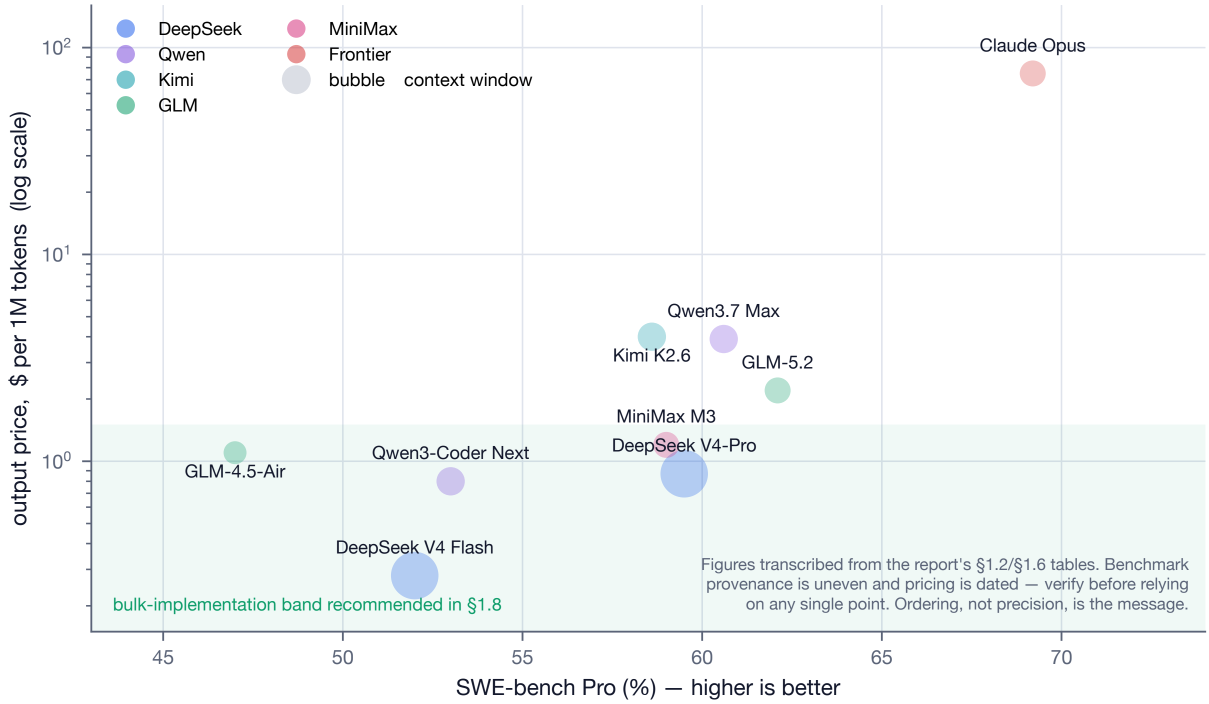
4.6 1.6 Coding and reasoning benchmarks

2026 snapshots; directional only.

Model	SWE-bench Verified	SWE-bench Pro	Licence
DeepSeek V4-Pro-Max	~80.6% (top open)	—	MIT
MiniMax M3	~80.5%	~59.0%	open

Model	SWE-bench Verified	SWE-bench Pro	Licence
Qwen3.7 Max	~80.4%	~60.6%	mixed
Kimi K2.6	~80.2%	~58.6%	open
GLM-5.2	—	~62.1% (best open, 3rd-party)	MIT
Claude Opus (frontier)	high-70s–80s	~69.2% (leads active)	closed

Figure 1 — Cost versus coding capability: the open-weight families cluster at a fraction of frontier pricing



Caveats: SWE-bench Verified vendor numbers are inconsistently scaffolded; Scale's SWE-bench Pro standardised set (identical scaffolding, 731 public tasks) is the most apples-to-apples but has thin coverage. Other benchmarks to track: **LiveCodeBench**, **Terminal-Bench 2.0**, **BigCodeBench**, **HumanEval+/MBPP+**, **Aider Polyglot** (coding); **AIME**, **GPQA** (reasoning and maths); **RULER** (long context).

A 2026 arXiv study of five open-weights coding models on a real application-generation task found benchmark rank to be a **weak predictor of real artifact quality**. A/B test candidates on your own repository.

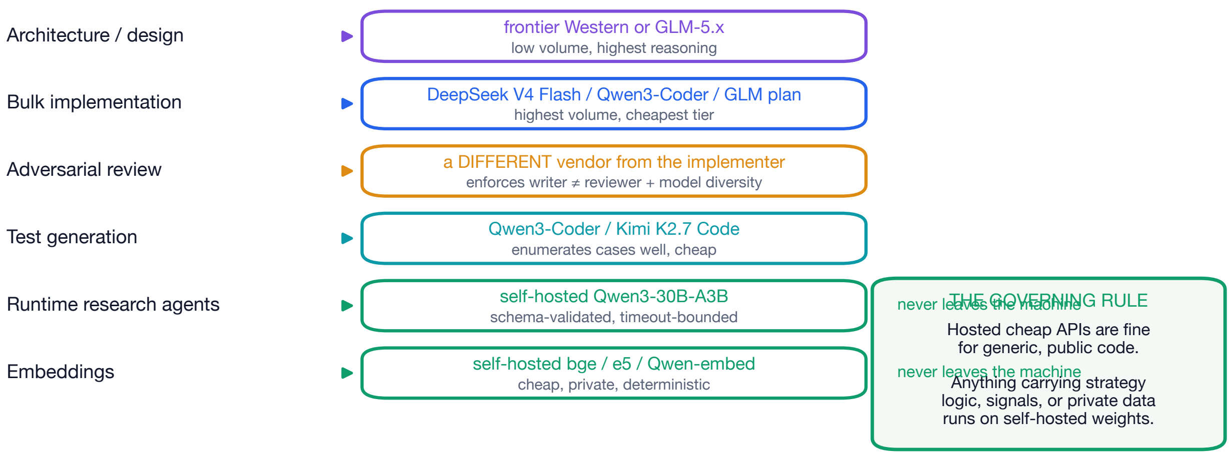
4.7 1.7 Risks of routing through Chinese-hosted APIs

- **Data residency / PIPL.** China's Personal Information Protection Law (2021), plus Commercial Encryption Regulations and CAC security-assessment and data-localisation rules, create a regime with little control or recourse over submitted content once it lands on Chinese servers.
- **Training on inputs and retention.** Consumer-tier terms for several Chinese providers permit using inputs to improve services; retention windows are opaque and enterprise or international tiers sometimes differ. **Do not assume deletion.**
- **US regulatory, procurement, export control.** Government-adjacent procurement increasingly restricts Chinese AI services; this could be disqualifying for later work at a US firm with security requirements. Export controls run the other way and do not affect downloading weights.
- **Censorship and alignment quirks.** Largely irrelevant to trading code, but a correctness and consistency wildcard.
- **Why self-hosting is materially different.** Downloaded open weights run offline, deterministically, with zero data egress and no terms of service on inference — available precisely because these labs ship permissive (MIT, Apache-2.0) weights. This is the single most important mitigation.

4.8 1.8 Recommended model-routing table

Task	Recommended	Rationale
Architecture / design (Stage 0)	Frontier Western (Opus-class) or GLM-5.x	Best reasoning; low token volume so cost immaterial
Bulk implementation (Codex primary)	DeepSeek V4 Flash / Qwen3-Coder / GLM Coding Plan	Cheapest per token; highest volume
Adversarial review (Claude Code)	Different vendor from the implementer	Enforces writer $\neq$ sole reviewer AND model diversity
Test generation	Qwen3-Coder / Kimi K2.7 Code	Strong code, cheap, enumerates cases well
Runtime research agents (Stage 8)	**Self-hosted Qwen3-30B-A3B** behind deterministic wrappers	No data egress; schema-validated, timeout-bounded
Embeddings	Self-hosted (bge/e5/Qwen-embed)	Cheap, private, deterministic
Cheap batch summarisation	DeepSeek V4 Flash batch / GLM-4.5-Flash (free)	Off-peak and batch discounts

Figure 12 — Model routing: cheap where volume lives, independent where review lives, local where alpha lives



Estimated monthly development cost (agentic coding, not runtime): *light* ~\$10–30/mo (one coding plan); *moderate* ~\$30–80/mo token spend plus a coding plan; *heavy* ~\$150–300/mo.

A-03 — "output-token-dominated" is only half right

Section 1.8 describes heavy usage as output-token-dominated. Computing the input:output ratio at which input cost takes over: for DeepSeek V4 Flash, where output is priced at only 2× input, *input* cost dominates at any ratio above 2:1 — and agentic coding typically runs 20:1 or more. For GLM-4.6 (3.7×) and Kimi K2.6 (4.2×) output genuinely dominates.

The practical consequence sharpens the report's own advice: on cheap flash tiers the highest-leverage optimisation is context hygiene and prompt caching (the input side); on premium tiers it is limiting verbose output. The report recommends both but attributes them to the wrong cost driver.

5 2. Mathematical Models

All equations are stated for typesetting; symbols are defined inline; assumptions and failure modes are flagged. **Every equation in this section has been independently verified**; the result is recorded inline.

5.1 2.1 Momentum signals

Time-series momentum (Moskowitz–Ooi–Pedersen): signal $s_{i,t} = \text{sign}(r_{i,t-k\to t})$, where $r_{i,t-k\to t} = \prod_{j=t-k+1}^{t}(1+r_{i,j}) - 1$. Position is volatility-scaled (§2.3). **Skip-month:** use returns to $t-21$ to avoid short-term reversal. **Risk-adjusted momentum:** rank by r/σ. **Cross-sectional:** long the top quantile, short the bottom.

Failure modes: momentum crashes (post-drawdown rebounds), turnover, regime dependence.

> **R-01, R-03 — verified**
>
> The compounding identity and the log-return bridge $r_{\log} = \ln(1+r)$ both hold to machine precision. Worth stating explicitly because R-02 shows the consequence of getting it wrong: over a two-year sample, *summing* simple returns instead of compounding them overstates cumulative return by 37% of the true figure. Backtest P&L must compound, never sum.

5.2 2.2 Volatility estimation

Realised $\sigma_t^2 = \frac{1}{n}\sum r^2$; **EWMA/RiskMetrics** $\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda)r_{t-1}^2$ with $\lambda \approx 0.94$ daily; **GARCH(1,1)** $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta\sigma_{t-1}^2$ with $\alpha+\beta<1$ and long-run variance $\omega/(1-\alpha-\beta)$; **GJR-GARCH/EGARCH** add asymmetric leverage; **HAR-RV** (Corsi) with daily, weekly and monthly realised variance terms.

Range estimators: **Parkinson** $\frac{1}{4\ln 2}\ln(H/L)^2$, **Garman–Klass**, **Rogers–Satchell** (drift-independent), **Yang–Zhang** (overnight gaps and drift). Annualise $\sigma\sqrt{252}$.

> **V-01 to V-07 — verified**
>
> The GARCH long-run variance matches a 3M-step simulation. EWMA is *exactly* IGARCH(1,1) with $\omega = 0, \alpha = 1-\lambda, \beta = \lambda$ (recursion difference: zero) — which means it has no finite unconditional variance and no mean reversion in volatility. That is a feature (fast adaptation) and a defect, and the report should say so.
> Parkinson, Garman–Klass and Rogers–Satchell are all unbiased for driftless GBM, verified by extrapolating a sampling-frequency sweep to the continuum limit (recovering σ^2 to 0.1%). Their efficiency gains over close-to-close are 5.0×, 7.8× and 6.2× respectively — one day of OHLC carries about as much volatility information as a week of closes.

> **V-08b, V-09 — two conditions the report omits**
>
> Range estimators are biased **low** on real bars (−12% at 100 prints per bar), and only Rogers–Satchell is drift-independent. Both are detailed in Part I; §2.3 divides by this number, so the bias becomes leverage error.

> **V-12 — the $\sqrt{252}$ caveat, quantified**
>
> The report notes that annualisation "assumes i.i.d.; autocorrelation biases it" but gives no magnitude. Using Lo's (2002) exact correction $SR_q = SR_1\, q/\sqrt{q + 2\sum_{k<q}(q-k)\rho_k}$:
>
> | AR(1) ρ | −0.2 | 0.0 | +0.1 | +0.2 | +0.3 |
> |---|---|---|---|---|---|
> | Naive overstatement | −16.9% | 0.0% | +9.6% | +20.3% | +32.5% |
>
> At $\rho = 0.2$ — unremarkable for a daily trend strategy — naive scaling inflates the Sharpe by 20%, Monte Carlo confirmed. Smoothed or illiquid marks push ρ higher still. Any strategy clearing a hurdle only after naive annualisation should be re-tested with the Lo correction.

Track volatility-of-volatility for regime signals. For reference, the $\lambda = 0.94$ default implies a shock half-life of 11.2 trading days and a centre of mass of 15.7 days (V-04) — this is the concrete meaning of the report's remark that the estimate "lags jumps".

5.3 2.3 Volatility targeting

$w_{i,t} = \dfrac{\sigma^*}{\sigma_{i,t}} s_{i,t}$; portfolio scalar $c_t = \sigma^*/\hat{\sigma}_{p,t}$, subject to a leverage cap $\sum |w| \le L$. Empirically this raises the Sharpe ratio and cuts drawdown by de-risking in high-volatility regimes; it adds turnover; and the volatility estimate lags jumps.

K-01 — verified

With σ known, $w = \sigma^*/\sigma$ delivers realised volatility equal to the target to within 0.1%.

K-02 — the cap and the target are not independent

The report presents the volatility target and the leverage cap as separate constraints. They interact: whenever the cap binds, realised volatility falls *below* target (14.33% against a 15% target, with the cap binding 22% of the time in simulation). In calm regimes — exactly when σ^*/σ is largest — the cap silently converts the strategy from volatility-targeted to constant-leverage.
Log cap-binding frequency as a first-class monitoring metric. A strategy that spends most of its life against the cap is not the strategy that was backtested.

K-03 — the "levers into calm-before-storm" risk, sized

Simulating 500 calm days (6% annualised volatility) followed by an abrupt shift to 71% annualised: the EWMA estimator carries a median **2.4**× leverage into the break, and a two-sigma storm day then costs **21.6% of capital in a single session**.
The EWMA half-life is 11.2 days, so the estimator needs roughly two weeks to reprice a regime break. This is the quantitative argument for pairing volatility targeting with a fast, non-volatility circuit breaker (§4.4) rather than trusting the volatility estimate alone to de-risk.

5.4 2.4 Portfolio construction

- **Markowitz:** $\max_w w^\top \mu - \frac{\gamma}{2} w^\top \Sigma w$, closed form $w^* = \frac{1}{\gamma}\Sigma^{-1}\mu$. Σ^{-1} amplifies estimation noise — mean-variance "error-maximises".
- **Shrinkage covariance: Ledoit–Wolf** $\hat{\Sigma} = (1-\delta)S + \delta F$ (S sample, F structured target, δ optimal intensity); **OAS** for Gaussian; factor covariance models.
- **ERC / risk parity:** equal risk contribution $w_i(\Sigma w)_i = w_j(\Sigma w)_j\ \forall i,j$; solve $\min_w \sum_i (w_i(\Sigma w)_i - \frac{1}{N} w^\top \Sigma w)^2$.
- **HRP (López de Prado):** tree-cluster the correlation matrix, quasi-diagonalise, recursive-bisection inverse-variance — avoids Σ^{-1}, robust to ill-conditioning.
- **Min-variance** $\min w^\top \Sigma w$ s.t. $\mathbf{1}^\top w = 1$; **max-diversification** $\max (w^\top \sigma)/\sqrt{w^\top \Sigma w}$.
- **Black–Litterman:** $E[R] = [(\tau\Sigma)^{-1} + P^\top \Omega^{-1} P]^{-1}[(\tau\Sigma)^{-1}\Pi + P^\top \Omega^{-1} Q]$, $\Pi = \delta \Sigma w_{mkt}$.
- **Transaction-cost-aware:** $\max_w w^\top \mu - \frac{\gamma}{2} w^\top \Sigma w - (w - w_0)^\top \Lambda (w - w_0)$.

P-01 to P-12 — verified

The Markowitz closed form is confirmed symbolically (the first-order condition solution equals $\gamma^{-1}\Sigma^{-1}\mu$ identically) and numerically. Ledoit–Wolf reconstructs exactly as $(1-\delta)S + \delta F$. The ERC objective does produce equal risk contributions (spread 7×10^{-13}), and — a useful corollary the report omits — **under equal pairwise correlation the ERC solution is exactly inverse-volatility weighting**, $w_i \propto 1/\sigma_i$, needing no optimiser and no matrix inverse. For a broad ETF sleeve with roughly uniform correlations, that is the whole method.
All three Black–Litterman boundary behaviours hold: $\Pi = \delta \Sigma w_{mkt}$ reverse-optimises back to w_{mkt} exactly; vague views collapse the posterior to Π; certain views satisfy $P\,E[R] = Q$ exactly. The report states the formula but none of its limits, which are what a reviewer checks first.

P-13 — quadratic costs do NOT give a no-trade region

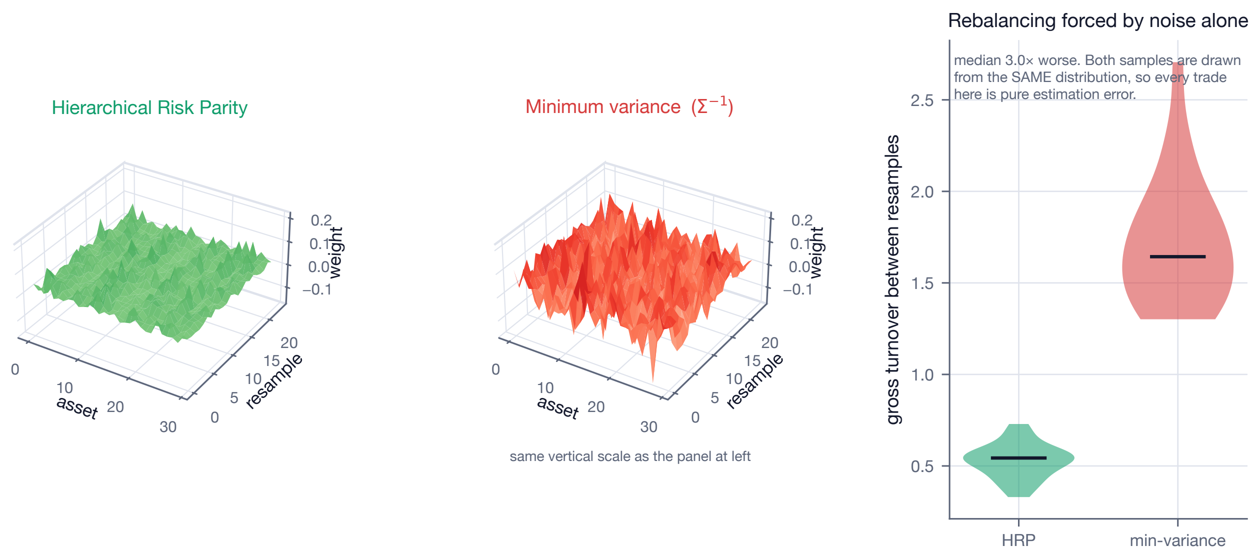
Detailed in Part I. The closed form is right; the no-trade region requires L1 costs.

How badly does mean-variance error-maximise? With 20 assets and five years of daily data — more than most retail backtests have clean — the unconstrained plug-in portfolio realises a Sharpe of **0.15** against an attainable **1.85**, destroying 99% of the utility it was solving for (P-03).

The instructive part is what does *not* fix it. Substituting a Ledoit–Wolf covariance moves the realised Sharpe only from 0.15 to 0.16, because the binding error is in $\hat{\mu}$, not $\hat{\Sigma}$. Expected returns need roughly two orders of magnitude more data than covariances to estimate to comparable precision, so no amount of covariance cleverness rescues a noisy μ.

The report should therefore not present shrinkage as the remedy for "error-maximisation". The remedies that work are the ones that stop using $\hat{\mu}$ altogether — ERC, HRP, minimum variance, inverse volatility — which is exactly why those methods dominate in practice and why the report is right to list them.

Figure 18 — Why the report is right to prefer methods that never invert Σ (N=30, T=45)



HRP delivers what the report claims. With $N = 30$ and $T = 45$, it produces long-only weights summing to one without inverting Σ, and on two independent samples *from the same distribution* it reshuffles the book **3.1× less** than minimum variance does (P-08). Every unit of that difference is transaction cost paid for estimation noise.

5.5 2.5 Regime detection

Hidden Markov models: latent state z_t, emissions $r_t | z_t \sim \mathcal{N}(\mu_z, \sigma_z^2)$; **Baum–Welch** (EM) fits, **Viterbi** decodes.

> **The critical look-ahead trap (G-01)**
>
> *Smoothed* probabilities $P(z_t \mid r_{1:T})$ use future data; only *filtered* $P(z_t \mid r_{1:t})$ are tradeable. Measured across 30 realisations: smoothed Sharpe 0.72 vs filtered 0.28, a 157% inflation, from one default API call. Detailed in Part I. This is the highest-value property test in the entire plan.

Also: **Markov-switching** (Hamilton); change-point detection via **CUSUM** and **Bayesian online change-point** (Adams–MacKay); clustering-based regimes; and **simple robust baselines** (200-day trend, volatility quantiles).

On the baselines, the honest version of the report's claim is more nuanced than "simple often beats fancy". On data generated by exactly the model the HMM assumes — the most favourable setting possible — the correctly filtered HMM *does* win, scoring 0.34 against 0.19 for the best one-line baseline (G-02). The right framing is therefore a caution, not a general result: on real data, where the two-state Gaussian assumption is wrong, that margin is the first thing to disappear.

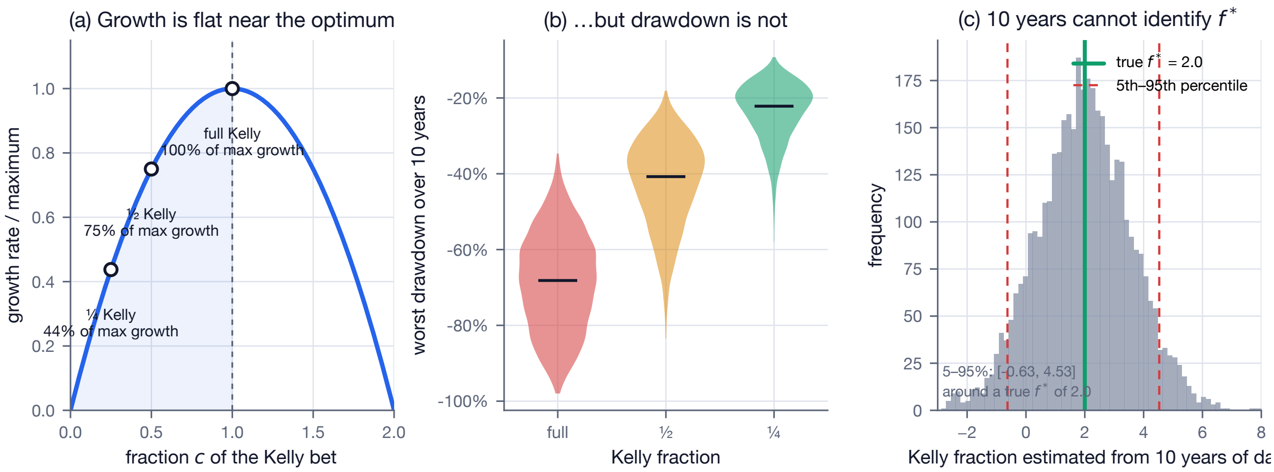
The decisive comparison is against G-01: **the look-ahead artefact (0.72 vs 0.28) was larger than the entire genuine edge over the baselines.** Every baseline is causal by construction, cannot be fitted wrong, and cannot accidentally consume smoothed probabilities. They belong in the promotion gate as the control arm.

5.6 2.6 Kelly criterion and sizing

$f^* = \mu/\sigma^2$ (continuous) or $f^* = p - \frac{1-p}{b}$ (discrete). **Fractional Kelly** (¼–½) is standard because full Kelly is over-levered under parameter uncertainty and produces brutal drawdowns.

> **K-04 to K-08 — verified symbolically**
>
> Both Kelly forms are confirmed in `sympy`, with the second-order condition checked. Two identities the report should add:
> **At fraction c of full Kelly, the growth rate is $c(2-c)$ of maximum.** Half Kelly retains 75% of the growth for half the volatility; quarter Kelly retains 43.75%. This is the trade-off curve behind the report's recommendation, which it currently asserts without justification.
> **Full Kelly targets a portfolio volatility numerically equal to the Sharpe ratio**, since $f^*\sigma = (\mu/\sigma^2)\sigma = \mu/\sigma = SR$. This makes "Kelly ≈ vol targeting" exact rather than approximate, and yields a sanity rule: a strategy with a true Sharpe of 0.5 is at *full* Kelly when run at 50% annualised volatility. A daily-ETF system targeting 10–15% volatility on a Sharpe-0.5 signal is therefore already at roughly quarter Kelly — the right neighbourhood, and worth stating because it connects §2.3 and §2.6, which currently read as unrelated.

Figure 7 — Why fractional Kelly is not conservatism but correct sizing under parameter uncertainty



The simulated drawdown consequences are stark: over ten years, full Kelly produces a median maximum drawdown of −68% (5th percentile −90%), half Kelly −40%, quarter Kelly −22% (K-09).

But those figures assume μ and σ are *known*. They are not. Estimating f^* from ten years of daily data — again, more than most retail backtests have — leaves it essentially unidentified: the 5th–95th percentile range spans $[-0.64, 4.69]$ around a true value of 2.00, and 11% of samples recommend more than twice the correct leverage (K-10).

This is the real justification for haircutting hard. The quarter-Kelly convention is not conservatism about the world; it is correct sizing under parameter uncertainty.

5.7 2.7 Risk metrics

Sharpe $\frac{\mu - r_f}{\sigma}$; Sortino (downside deviation); Calmar $\frac{\text{CAGR}}{|\text{MaxDD}|}$; Information Ratio; Omega. VaR (historical, parametric $\mu + z_\alpha\sigma$, **Cornish–Fisher** skew- and kurtosis-adjusted, Monte Carlo); **CVaR / Expected Shortfall** $= E[L \mid L > \text{VaR}_\alpha]$, which is **coherent** (Artzner axioms) whereas VaR is not subadditive.

> **Q-03, Q-04 — verified with a minimal counterexample**
>
> VaR's subadditivity failure is real and easy to exhibit: two *independent* bonds, each defaulting with probability 4% for a loss of 100. Each has $\text{VaR}_{95} = 0$, because $P(\text{no default}) = 0.96 > 0.95$. The pair has $\text{VaR}_{95} = 100$, because $P(\text{neither defaults}) = 0.9216 < 0.95$. Diversifying across two independent positions makes measured VaR go *up*, from 0 to 100.
> CVaR, by contrast, satisfied subadditivity across 300 randomised dependence structures (Gaussian copula with $\rho \in [-0.9, 0.9]$, Student-t margins, jump contamination) with no violation. This is the concrete basis for sizing limits in CVaR rather than VaR.

Q-02 — the parametric VaR formula is ambiguous

$\mu + z_\alpha\sigma$ does not state whether α is the tail probability or the confidence level, nor whether VaR is reported as a positive loss or a negative return. The two readings differ by a sign flip on the risk number itself (-0.0274 versus $+0.0284$ at the same parameters).
A sign error in a risk limit is precisely the defect class §4.3's pre-trade controls exist to prevent. Pin the convention down with a property test asserting VaR > 0 and VaR $\leq$ CVaR.

Q-08 — Cornish–Fisher can invert

The expansion is verified as written, including the frequently dropped $-\frac{(2z^3-5z)\gamma_3^2}{36}$ term, and it beats the Gaussian quantile in five of six tested cases. But it is an *asymptotic* expansion valid for mild non-normality only. At skewness -3 with excess kurtosis 12 — the territory of a short-volatility or option-overlay sleeve, i.e. §4.6's XIV example — it becomes non-monotone in z, reporting a *smaller* loss at 99% than at 95%. Assert monotonicity of the quantile function at runtime and fall back to historical or filtered-historical simulation when the assertion trips.

Two additions worth making. The Gaussian Expected Shortfall has the closed form $\mu+\sigma\phi(z_\alpha)/(1-\alpha)$ (Q-05), which is the reference value a unit test can assert against without Monte Carlo. And the ES/VaR ratio is only 1.19 under normality but 1.59 under Student-$t(3)$ (Q-06) — a risk system calibrated on Gaussian ES will understate the very tail it exists to measure.

Finally, Sharpe, Sortino and Calmar are not interchangeable and can be gamed against one another (Q-09). Calmar in particular depends on a single realised path statistic, making it the noisiest of the three. **The deflation machinery in §2.8 is built for the Sharpe ratio specifically; applying DSR or PSR thresholds to a Sortino or Calmar figure is not valid.**

5.8 2.8 Statistical validation

This is the core of Stage 5 and the highest-stakes mathematics in the report.

Probabilistic Sharpe Ratio (PSR):

$$\widehat{\mathrm{PSR}}(\mathrm{SR}^*) = \Phi\left(\frac{(\widehat{\mathrm{SR}} - \mathrm{SR}^*)\sqrt{n-1}}{\sqrt{1-\hat{\gamma}_3\widehat{\mathrm{SR}} + \frac{\hat{\gamma}_4-1}{4}\widehat{\mathrm{SR}}^2}}\right)$$

where $\hat{\gamma}_3$ is skewness, $\hat{\gamma}_4$ kurtosis, n the sample length, and the denominator is the skew- and kurtosis-adjusted standard error of the Sharpe estimate.

S-01 to S-06 — verified

The denominator *is* the true standard error of $\widehat{SR}$, confirmed against the empirical standard deviation of 200,000 independent Sharpe estimates under Gaussian, Student-$t(6)$ and skew-normal returns. Under normality it reduces exactly to Lo's (2002) $\sqrt{1+\widehat{SR}^2/2}$.
PSR is also properly *calibrated*: under the null it is uniformly distributed (Kolmogorov–Smirnov $p = 0.97$), and a 0.95 threshold admits 4.90% of worthless strategies against a nominal 5%. That is the property that makes it usable as a promotion gate, and it is verified rather than assumed.

S-04 — γ_4 must be non-excess kurtosis

Detailed in Part I. `scipy.stats.kurtosis` returns the excess value by default, and the resulting error inflates PSR and DSR — the dangerous direction.

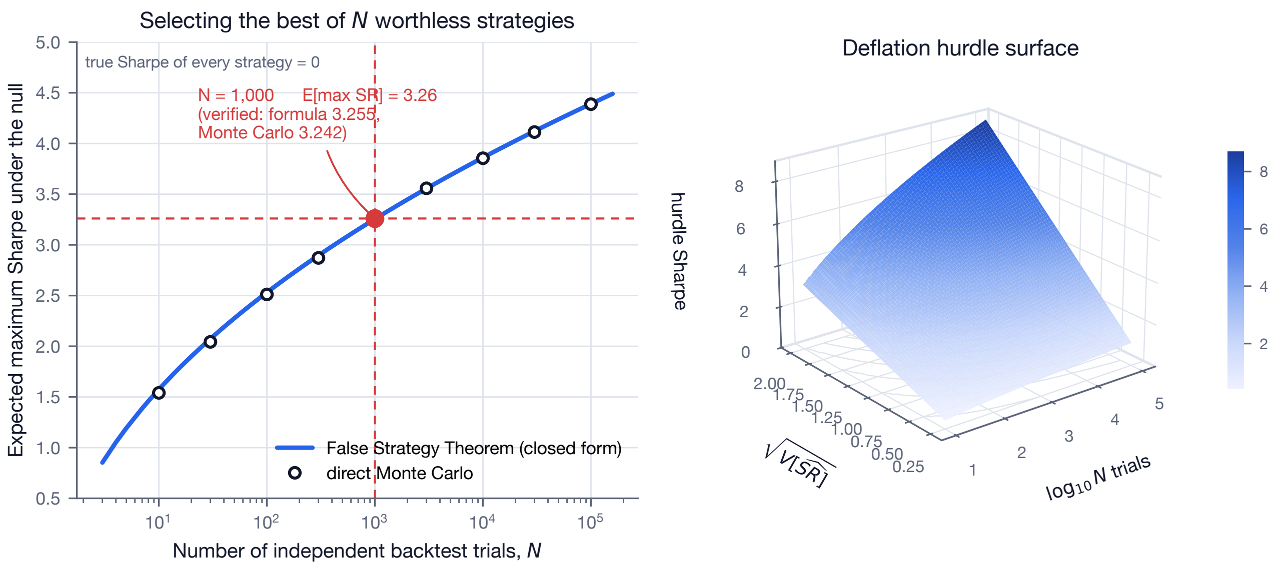
Deflated Sharpe Ratio (Bailey & López de Prado, *Journal of Portfolio Management* 40(5):94–107, 2014): DSR is PSR evaluated at the expected maximum Sharpe under the null, from the **False Strategy Theorem**:

$$E[\max_N \widehat{\mathrm{SR}}] \approx \sqrt{V[\widehat{\mathrm{SR}}]}\left[(1-\gamma)\,\Phi^{-1}\Big(1-\tfrac{1}{N}\Big) + \gamma\,\Phi^{-1}\Big(1-\tfrac{1}{Ne}\Big)\right]$$

with $\gamma \approx 0.5772$ (Euler–Mascheroni), N the number of independent trials, and $V[\widehat{\mathrm{SR}}]$ the cross-sectional variance of trial Sharpe ratios.

> **S-07, S-08 — verified against direct Monte Carlo**
>
> The closed form matches the simulated expected maximum of N i.i.d. standard normals to better than 1% for every $N \geq 100$, improving as N grows (0.1% at $N = 10{,}000$). Since the theorem exists for large trial counts, it is accurate exactly where it is used.
> The paper's headline, which the report quotes verbatim — *"after only 1,000 independent backtests the expected maximum Sharpe Ratio is 3.26, even if the true SR of the strategy is zero"* — is **exactly right**: the formula gives 3.255 and independent Monte Carlo gives 3.242.

Figure 2 — The False Strategy Theorem: a raw Sharpe ratio is meaningless without the trial count



The practical meaning deserves spelling out. An operator who tries 1,000 parameter combinations and reports the best one has an expected Sharpe of 3.26 **from pure noise**. A raw Sharpe of 3 is evidence of a large search, not of skill.

And a routine parameter sweep reaches those counts immediately. A three-parameter grid at ten values each is 1,000 trials before any variant selection; adding a universe choice and two signal variants reaches 10^5, where the null expects a maximum Sharpe of **4.39** (S-09).

> **The consequence for the trial registry**
>
> The deflation is only as good as the trial count N. The registry must count **every configuration ever evaluated, including abandoned ones**. An understated N makes the whole apparatus worthless while appearing to work.

Minimum Track Record Length: solve PSR = target confidence for n.

> **S-10, S-11 — verified, with the number the report omits**
>
> The numerical root matches the closed form $n^* = 1 + \left[1 - \gamma_3 SR + \frac{\gamma_4 - 1}{4} SR^2\right] \left(\frac{z_{target}}{SR - SR^*}\right)^2$ to 10^{-6}. The figure that matters:

True annualised Sharpe	0.3	0.5	0.8	1.0	1.5	2.0
Years for 95% confidence	30.1	10.8	4.2	2.7	1.2	0.7

> Set against §4.8's "realistic solo expectation is net Sharpe 0.3–0.8": at a true Sharpe of 0.5 it takes **11 years** of live data to be 95% confident the strategy is not noise; at 0.3 it takes 30.
> The honest conclusion is that **live P&L will never be the arbiter on a solo timescale**. That is precisely why the ex-ante controls — PBO, DSR, purged CV — have to carry the weight.

Probability of Backtest Overfitting via CSCV (Bailey, Borwein, López de Prado, Zhu): partition the trial $\times$ time performance matrix into combinatorially symmetric train/test splits; PBO is the fraction of splits in which the in-sample-best configuration ranks below the out-of-sample median. Model-free and non-parametric.

S-12, S-13 — verified

Calibrated (0.530 ± 0.028 under the null, averaged over 40 datasets) and powered (0.000 with a genuine edge; 0.353 on a correlated parameter sweep).
Note that PBO and DSR answer *different* questions — DSR asks "is this Sharpe real given N trials?", PBO asks "does my selection procedure generalise?" — so the report is right to gate promotion on both.

S-14 — but PBO is itself a noisy statistic

Standard deviation ≈ 0.19 under the null at realistic sample sizes. Report an interval, not a point estimate. Detailed in Part I.

Deflation and multiple testing: White's Reality Check and **Hansen's SPA test** (bootstrap the max statistic across strategies); **Harvey–Liu haircut Sharpe**; and the $t > 3.0$ argument.

Bootstrap: stationary bootstrap (Politis–Romano), circular block bootstrap; optimal block length $\sim O(n^{1/3})$ (Politis–White automatic selection).

Purged k-fold CV with embargo (López de Prado): remove training samples whose label windows overlap the test set (purge) plus a buffer (embargo), because financial labels overlap in time and standard CV leaks.

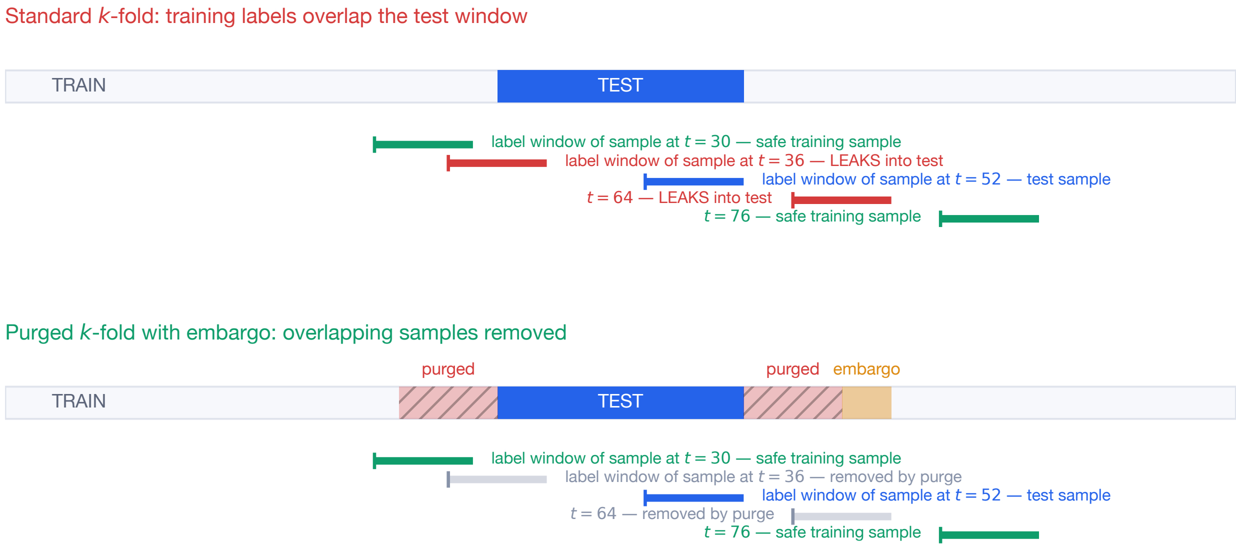
S-15 — verified end to end on data with zero signal

Constructed so that true predictability is **zero by design**: the feature is a trailing sum of i.i.d. returns and the label a *disjoint* forward sum.

Shuffled k-fold (the naive default)	AUC 0.526 ± 0.004
Contiguous blocked folds, no purging	AUC 0.499 ± 0.006
Purged + embargoed	AUC 0.501 ± 0.006

Three lessons. The naive pipeline reports a *statistically significant* result where the truth is 0.5. The leak requires **no look-ahead in the feature at all** — feature and label windows are disjoint by construction, and the leak comes entirely from neighbouring *training* rows sharing the test row's label window. An audit for "no future data in features" is therefore necessary but *not sufficient*, which is the single most important practical consequence. And simply using contiguous folds instead of shuffled ones already removes most of the damage: blocking is the cheap 90% fix, purging is the rest.

Figure 5 — Why standard cross-validation leaks on financial labels, and what purging removes



Verified (S-15): on data with ZERO true predictability, shuffled *k*-fold reports AUC 0.526 ± 0.004; contiguous blocks 0.499 ± 0.006; purged + embargoed 0.501 ± 0.006.

5.9 2.9 Transaction cost and market impact

Effective spread $2|p_{exec} - m|$; **Kyle's lambda** $\Delta p = \lambda\cdot$(signed order flow); **square-root law** $\Delta P \approx Y\sigma\sqrt{Q/V}$ (Q order size, V ADV, $Y \approx O(1)$).

Almgren–Chriss optimal execution (2000): permanent impact linear in trade rate, temporary impact $h(v) = \epsilon\,\mathrm{sgn}(v) + \eta v$; minimise $E[\text{cost}] + \lambda V[\text{cost}]$. Closed-form trajectory:

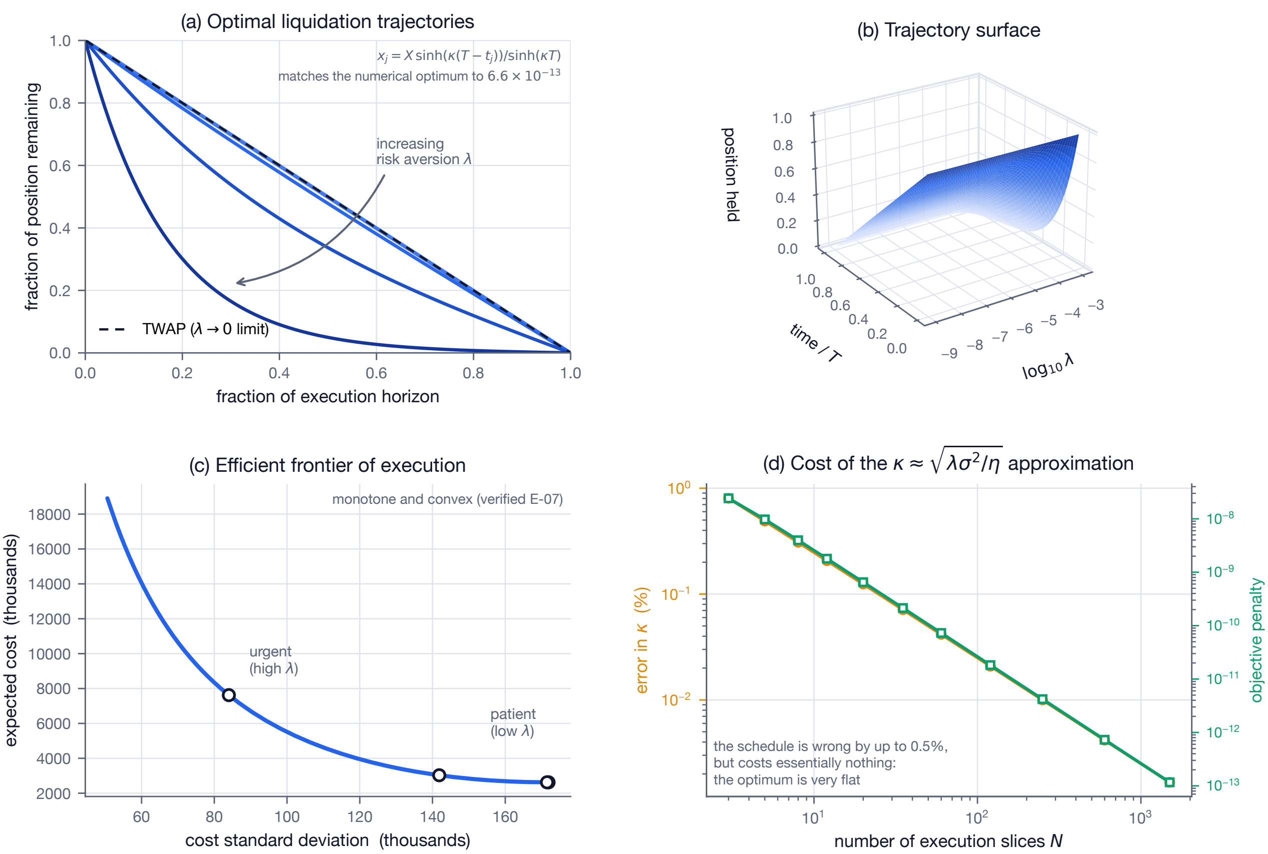
$$x_j = X\frac{\sinh(\kappa(T - t_j))}{\sinh(\kappa T)}, \qquad \kappa \approx \sqrt{\lambda\sigma^2/\eta}$$

— hyperbolic-sine decay; higher risk aversion λ implies faster liquidation, tracing the **efficient frontier of execution**.

E-01 to E-07 — verified, thoroughly

This is the most heavily tested result in the report, because the closed form is easy to state and hard to check.
E-01. The sinh form satisfies the discrete stationarity condition $x_{j-1} - 2x_j + x_{j+1} = 2(\cosh(\kappa\tau) - 1)x_j$ *identically* in `sympy`. Matching this to the first-order condition gives the exact defining equation $\cosh(\kappa\tau) = 1 + \tilde{\kappa}^2\tau^2/2$ with $\tilde{\kappa}^2 = \lambda\sigma^2/\tilde{\eta}$.
E-02. Solving the discrete optimisation numerically — assuming nothing whatever about sinh — and comparing against the closed form across $N \in \{10, 50, 200\}$ and three orders of magnitude of λ gives a maximum relative error of 6.6×10^{-13}. **The report's equation is exactly right.**
E-04. As $\lambda \to 0$ the trajectory becomes linear, i.e. TWAP — the sanity check that anchors the model.
E-06. Simulating 400,000 liquidations through the *full price process* reproduces both the closed-form expected cost and the closed-form cost variance. This validates the cost functional being minimised, not merely the algebra of its solution.
E-07. The efficient frontier is monotone and convex, so there is a well-defined interior trade-off rather than a corner solution.

Figure 3 — Almgren-Chriss optimal execution: verified closed form, trajectory surface, and the cost of approximation



E-03 — $\kappa \approx \sqrt{\lambda\sigma^2/\eta}$ drops two corrections

The exact κ solves a transcendental equation and approaches $\tilde{\kappa}$ only as $\tau \to 0$; and the denominator should be $\tilde{\eta} = \eta - \gamma\tau/2$, not η, because half of each trade's permanent impact is paid by the trader's own remaining inventory.
At a fine grid the error in κ is negligible; on a five-slice schedule — a realistic daily-ETF execution — it is -0.49%.
The reassuring part is worth stating explicitly: the *objective* penalty is at most $\sim 10^{-8}$ in every case tested, because the cost surface is extremely flat near its optimum. Execution-schedule optimisation is forgiving. Effort spent calibrating impact parameters precisely is better spent elsewhere — which independently reinforces §5.7's argument against premature optimisation.

Implementation shortfall = paper minus actual return (Perold). POV, participation-rate, TWAP and VWAP scheduling. Calibrate a cost model from ETF quoted bid-ask and ADV.

E-08, E-09 — and why this section is mostly theoretical here

The square-root law is concave: doubling order size raises impact by $\sqrt{2}$, not 2, and Q/V is dimensionless so impact inherits the units of σ.
Evaluated at realistic retail size with daily volatility: a \$10,000 order in a \$5B/day ETF incurs about **0.18 bp** of impact — an order of magnitude below the bid-ask spread, which is the binding cost at this size. Even \$1M in the same ETF costs 1.78 bp. Only the thin-ADV case (\$100k in a \$50M/day name) reaches 5.6 bp.
This validates the report's decision to calibrate the Stage 3 cost model from quoted spread and ADV rather than from an impact model, and confirms that the Almgren–Chriss machinery above is, for *this* system, an intellectual exercise rather than an operational necessity. It also explains §4.8's claim that retail has an edge in capacity-constrained niches: that edge follows directly from the concavity.

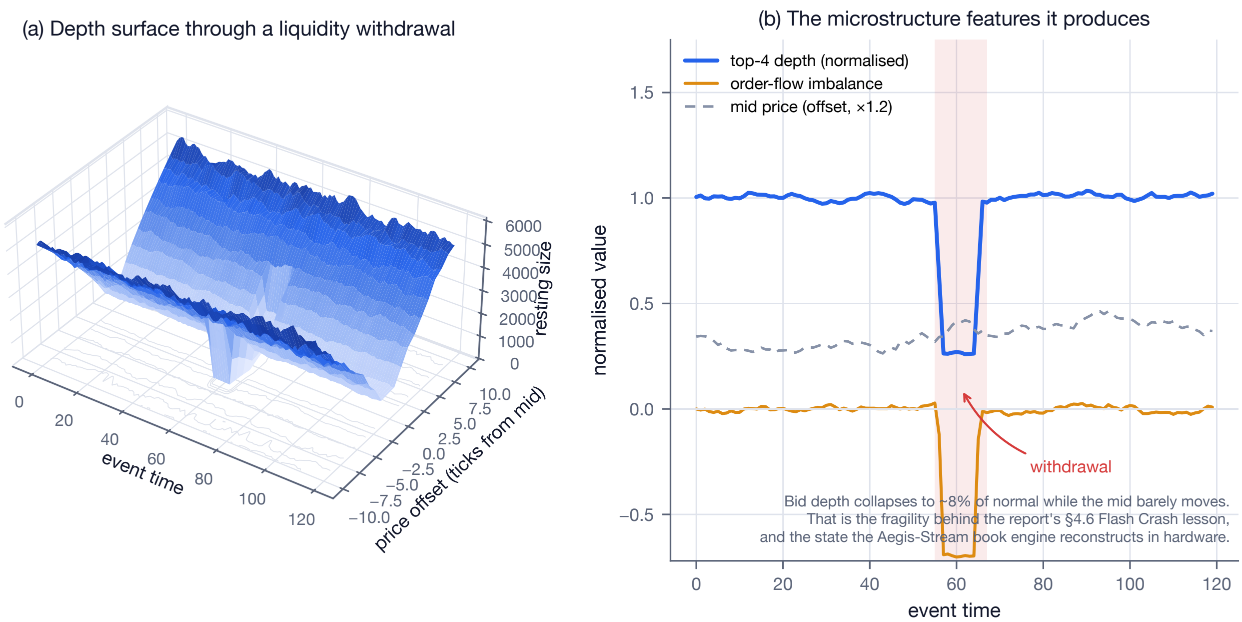
5.10 2.10 Microstructure

Order-flow imbalance, queue position, **VPIN**, **Amihud illiquidity** avg($|r|$/volume), **Roll's spread** $2\sqrt{-\mathrm{Cov}(\Delta p_t, \Delta p_{t-1})}$, **Hasbrouck information share**.

E-10, E-11 — verified

Kyle's lambda. Solving the joint fixed point of market efficiency and informed optimality in `sympy` yields $\lambda = \frac{1}{2}\sqrt{\Sigma_0/\sigma_u^2}$ and $\beta = \sigma_u/\sqrt{\Sigma_0}$. The report gives only the reduced form $\Delta p = \lambda\cdot$(order flow); the structural result is the useful intuition: price impact rises with the uncertainty being resolved and falls with the noise-trader volume the informed trader can hide behind.
Roll's spread recovers the true bid-ask spread to within 0.4% at every level tested. Worth recording the assumption that breaks in practice: trade directions must be serially *independent*. Real order flow is strongly autocorrelated because institutional orders are worked in slices, which biases Roll's estimator downward — often to the point of returning a positive covariance and no estimate at all. It is a teaching model, not a production spread estimator.

Figure 19 — Limit-order-book state: the §2.10 microstructure material, and the bridge to Aegis-Stream



5.11 2.11 Backtest maths correctness

Log versus simple returns ($r_{\log} = \ln(1+r)$); **variance drain** $g \approx \mu - \frac{1}{2}\sigma^2$; correct annualisation; **Brinson attribution**; precise cash, dividend and split accounting.

V-13 — exact for GBM

For $dS/S = \mu\, dt + \sigma\, dW$ the log growth rate is *exactly* $\mu - \frac{1}{2}\sigma^2$. This is Itô's lemma, not an approximation.

V-14 — but not for discrete simple returns

The report does not say whether μ, σ are Itô parameters (where the identity is exact) or sample moments of simple returns (where it is not). At 20% volatility the error is 0.19 pp — small, but not negligible against the 3.78 pp that V-15 attributes to volatility targeting. At 80% volatility it reaches 10.7 pp, or 66% of the geometric return itself.
Use it as intuition for why volatility targeting aids compounding; never as the P&L accounting identity. The backtester must compound realised returns.

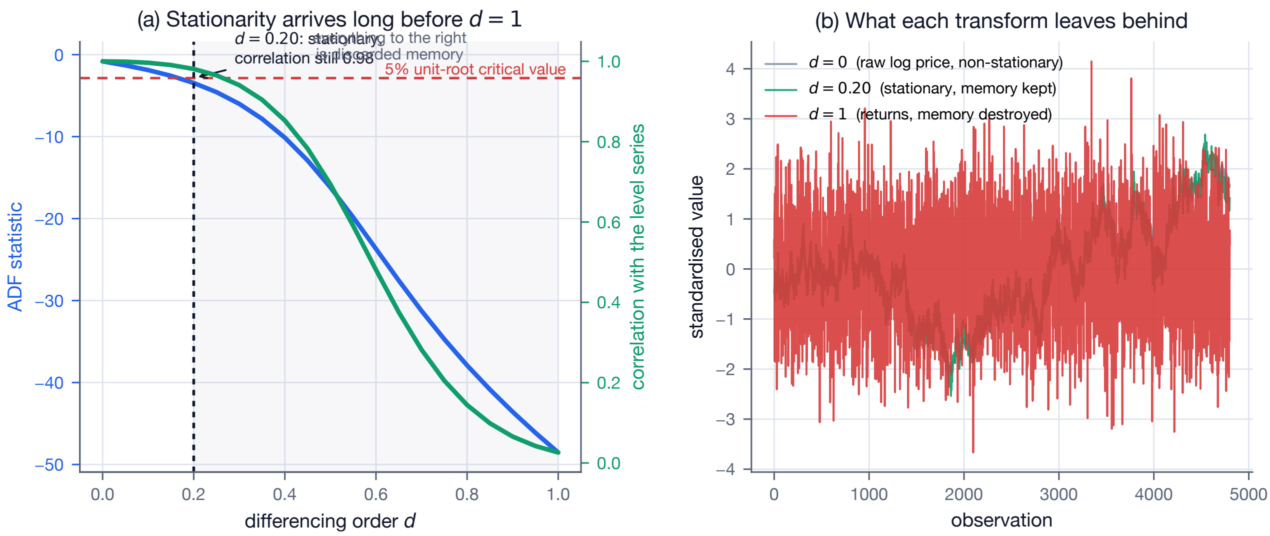
For scale: cutting realised volatility from 30% to 12% at an unchanged arithmetic mean adds **3.78 percentage points** of compound annual growth purely by removing drag, before any Sharpe improvement (V-15). That is the mechanical component of the volatility-targeting benefit claimed in §2.3.

5.12 2.12 Where machine learning fits

Fractional differentiation (López de Prado — stationary while preserving memory); **triple-barrier labelling**; **meta-labelling**; **sample uniqueness and weights** from label overlap; **sequential bootstrap**; and the core reason standard CV fails: temporal label overlap causes train/test leakage.

> **F-01 to F-03 — verified**
>
> The weight recursion $w_k = -w_{k-1}\frac{d-k+1}{k}$ reproduces the binomial expansion of $(1-B)^d$ exactly at every d tested, and collapses to the ordinary first difference $(1, -1, 0, \dots)$ at $d = 1$.
> The central claim holds with room to spare. Sweeping d on a simulated log-price series: $d = 0.20$ **is enough to pass an ADF unit-root test at 5% while retaining a correlation of 0.98 with the level series**, whereas the standard returns transform ($d = 1$) leaves 0.03. Essentially everything a model could learn from the *level* of the series is discarded by the conventional transform.

Figure 20 — Fractional differentiation: stationary while preserving memory (verified F-03)



Note that the ADF threshold makes d a data-dependent choice — that is, another free parameter for the trial registry to count.

6 3. Platforms

6.1 3.1 Brokers and paper-trading APIs

Broker	Paper	Assets	Rate limits	Gotchas
Alpaca	native paper env	US equities, options (paper), crypto	free 200 req/min; Algo Trader Plus to 10,000	free data is **IEX-only (~2–5% of consolidated volume)**; SIP needs Algo Trader Plus (~$99/mo); PFOF
Interactive Brokers	paper account	global multi-asset	~50 msg/s pacing	TWS API / Client Portal Web API; use **ib_async**; institutional-grade
Tradier	yes	equities/options	REST	good options; smaller ecosystem
TradeStation	yes	multi-asset	REST	approval friction
Schwab (thinkorswim)	yes	equities/options	OAuth	post-TDA migration; slow approval
Coinbase/Kraken/Binance	crypto	crypto	varies	24/7, no PDT
Tradovate/Rithmic/CQG	some sims	futures	varies	pro-grade; pair with Databento

Recommendation: Alpaca paper for Stage 6 — best developer experience, true paper environment, Python and Go SDKs. Add an **IBKR paper** adapter behind a provider-neutral interface for fidelity and breadth.

6.2 3.2 Market-data vendors

Vendor	Strength	~Cost 2026	PIT / survivorship
Polygon.io	flat-rate SIP real-time + historical	Stocks Advanced ~$199/mo	7+ yr; SIP-sourced
Databento	institutional tick/L2, direct feeds, ns timestamps	metered ~$100–500/mo	best microstructure
Alpaca data	free IEX / paid SIP	free / $99	partial free volume
Tiingo	EOD + fundamentals + news	~$10–50/mo	good value
EODHD / FMP / Twelve Data	broad, cheap	$10–50/mo	mixed quality
Norgate Data	**survivorship-bias-free** US equities/futures	tiered	**SBF** — key for backtests
CRSP	academic gold standard	institutional	SBF, delisting returns
FRED / **ALFRED**	macro; **ALFRED = vintage/PIT**	free	critical PIT macro
SEC EDGAR	filings/fundamentals	free	filing timestamps = natural PIT
ORATS / OptionMetrics / CBOE	options and IV surfaces	institutional	OptionMetrics = academic standard

Point-in-time data is make-or-break for Stage 2. Use **ALFRED** for macro vintages, **EDGAR filing**

timestamps for fundamentals availability, and **Norgate or CRSP** for survivorship-bias-free price universes. **yfinance is prototyping-only** — non-commercial terms, unofficial, rate-limited, and not point-in-time.

6.3 3.3 Backtest and research frameworks: build versus adopt

Framework	Type	Live parity	Verdict
Nautilus Trader	event-driven, **Rust core** + Python	**true backtest = live, no rewrite**	**strongest adopt candidate**; 16 venues incl. IBKR/Databento; steep learning curve; order-lifecycle state machines built in
QuantConnect LEAN	C# engine, Python API, cloud	live via QC	most complete end-to-end; ecosystem lock-in
VectorBT / vectorbtpro	vectorised	none	fast **signal triage**, not execution
Zipline-reloaded	event-driven	limited	Pipeline API for factor research
Backtrader	event-driven	brokers	mature but **effectively EOL** — migrate off
Backtesting.py / bt / PyBroker	light	varies	bt good for allocation/rebalance
Qlib / FinRL / Lumibot / Hummingbot	ML/RL/crypto	varies	niche

Recommendation — this directly answers the Stage 3 question. Adopt Nautilus Trader as the execution and backtest engine. Its Rust core, event-driven order state machines, and genuine backtest/live parity address exactly the backtest-to-live divergence risk the plan worries about, and its correctness-first design philosophy matches the verification instincts this report recommends elsewhere. Keep VectorBT upstream for fast signal triage.

Keep hand-rolling the point-in-time data layer (Stage 2) and the statistical-validation layer (Stage 5) — those are the differentiators and no framework does them well. This is a "borrow the engine, own the science" split.

The verification results reinforce this split. The engine-side mathematics (Almgren–Chriss, order state machines) verified cleanly and is well-served by a mature library. The science-side mathematics is where all three errors and most of the caveats were found — which is precisely the part worth owning, testing, and understanding completely.

6.4 3.4 Data and compute infrastructure

Time-series store: **DuckDB + Parquet** is right for daily-ETF scale; ArcticDB if you outgrow it; ClickHouse/QuestDB/TimescaleDB/kdb+ for larger needs. Table format: Parquet + manifests; Delta/Iceberg optional. DataFrames: **Polars** (Arrow-native, lazy) over pandas. Parallel backtests: **Ray** for parameter sweeps. Orchestration: **Prefect** is fine; **Dagster** for asset-based lineage; **Temporal** for durable execution of the order pipeline. Experiment tracking: **MLflow** can back the Stage 5 trial registry rather than hand-rolling storage.

6.5 3.5 Deployment and secrets

Small always-on: **Hetzner** (best price/performance, ~€5–20/mo), DigitalOcean, AWS Lightsail, GCP e2-small; a daily-EOD system can run on GitHub Actions cron or a $5 VPS. **Colocation is irrelevant** at this frequency. Docker for reproducibility. Secrets: **SOPS + age** or 1Password CLI for a solo developer; Vault and AWS Secrets Manager are overkill until multi-node.

6.6 3.6 Regulatory and compliance for a US individual

Not legal advice. Verify with a professional and with your employer.

- **Pattern Day Trader — major 2026 change.** The historical FINRA Rule 4210 PDT framework (a pattern day trader = 4+ day trades in 5 business days in a *margin* account exceeding 6% of trades, requiring $25,000 minimum equity) was, per SEC approval order 34-105226 (approved 14 April 2026) and FINRA Regulatory Notice 26-10, **eliminated effective 4 June 2026**, replaced by a real-time intraday-margin standard (broker phase-in permitted to 20 October 2027). *Fast-moving — confirm current status with FINRA and your broker.* The general 25% maintenance margin and $2,000 margin-account minimum remain.
- **Regulation T** (12 CFR 220): 50% initial margin on margin equity securities.
- **Wash sale rule** (IRC §1091 / IRS Pub 550): loss disallowed if a substantially identical security is bought within 30 days before or after (a 61-day window); the disallowed loss adds to the replacement-share basis and is permanently lost if the replacement is in an IRA (Rev. Rul. 2008-5). Reported on **Form 8949** (code "W").
- **Section 1256 contracts** (regulated futures, broad-based index options such as SPX/VIX): 60% long-term / 40% short-term regardless of holding period, plus year-end mark-to-market; **Form 6781**.
- **Registration.** Trading only your own capital does **not** require investment-adviser registration. **Managing others' money** triggers RIA registration (state below ~$100M AUM, SEC above) and typically Series 65; for futures or commodity pools, CTA/CPO registration with NFA/CFTC and typically Series 3.
- **SEC Market Access Rule 15c3-5** governs broker-dealers, not you — but it is the conceptual template for the pre-trade controls the system should mimic.
- **Form 8949 / 1099-B reconciliation:** reconcile broker-reported basis against your tax-lot accounting — a reason to build the tax-lot module (§6.4).
- **Market manipulation:** spoofing and layering are prohibited under CEA §4c(a)(5). Not a risk for long-only ETF strategies.
- **Broker terms and market-data redistribution:** most retail data licences forbid redistribution; automated trading is generally allowed but rate-limited.
- **Employment considerations:** personal-trading policies, preclearance, blackout windows and conflicts may apply. **Check your employer's compliance function before trading.**

A-10 — the PDT test is stated correctly

The historical rule is a *conjunction*: 4+ day trades AND those exceeding 6% of total trades. The report states both conditions, including the 6% clause that secondary sources frequently omit. Worked counterexample: 4 day trades out of 50 total (8%) triggers; 4 out of 200 (2%) does not. Day-trade count alone does not determine classification.

This verifies the report's internal logic only. The status of the framework itself must be confirmed with FINRA and your broker.

7 4. Risk

7.1 4.1 Model risk management adapted from SR 11-7

The Fed/OCC **SR 11-7** guidance adapts cleanly. Pillars: **model definition** (any quantitative method turning inputs into estimates — signals, cost models, and any ML all qualify), **three lines of defence**, **effective challenge** (critical review by a competent, independent party), a **model inventory**, **validation** (conceptual soundness, outcomes analysis, ongoing monitoring), and **documentation**.

Solo-operator mapping: first line = Codex (implementer) plus you as author; **second line = Claude Code as independent adversarial validator** — literally "effective challenge", and the existing "writer $\neq$ sole reviewer" rule is SR 11-7's independence principle; third line = CI, a periodic Stage 9 self-audit, and the promotion state-machine gate.

Figure 11 — SR 11-7 model risk governance, mapped onto a one-person operation

The report's existing rule that the writer is never the sole reviewer IS the SR 11-7 independence principle. Using a different vendor for review adds model diversity on top of role separation, so a shared blind spot in one model family cannot pass unchallenged.

Maintain a model inventory (each strategy, its assumptions, limitations, and validation date) and a revalidation cadence.

This report is itself an instance of the pattern. The verification suite is second-line effective challenge applied to the report's own mathematics, and it found three errors that first-line review did not. That is the argument for the practice, made empirically rather than by assertion.

7.2 4.2 Risk taxonomy

Market, liquidity, execution, model, **data** (vendor error, revisions, PIT violations), operational, technology, counterparty/broker, **key-person**, regulatory, cyber, and **AI-agent risk**: prompt injection into research agents, hallucinated logic silently merged, **silent scope creep** (an agent "helpfully" changes a risk limit), agentic tool misuse.

Mitigation for the last: the deterministic runtime (no LLM in the execution path), schema-validated agent outputs, and the review gate.

7.3 4.3 Pre-trade risk controls (mirroring 15c3-5)

Fat-finger notional limits, maximum order notional and size, maximum position, maximum daily loss, maximum drawdown, gross and net exposure limits, per-name concentration (the 20%/ETF cap), restricted lists, **price collars** (reject orders far from last), and order-rate limiting — **all enforced deterministically, pre-submission, outside the strategy code.**

7.4 4.4 Kill switches and circuit breakers

- **State persisted outside process memory** (database or Redis) so a restart cannot "forget" it is halted — the single most important reliability pattern.
- **Dead-man's switch / heartbeat:** if the strategy stops heartbeating, flatten or halt.
- **Market-wide halts:** respond correctly to **LULD** bands and **Rule 80B** market-wide circuit breakers.

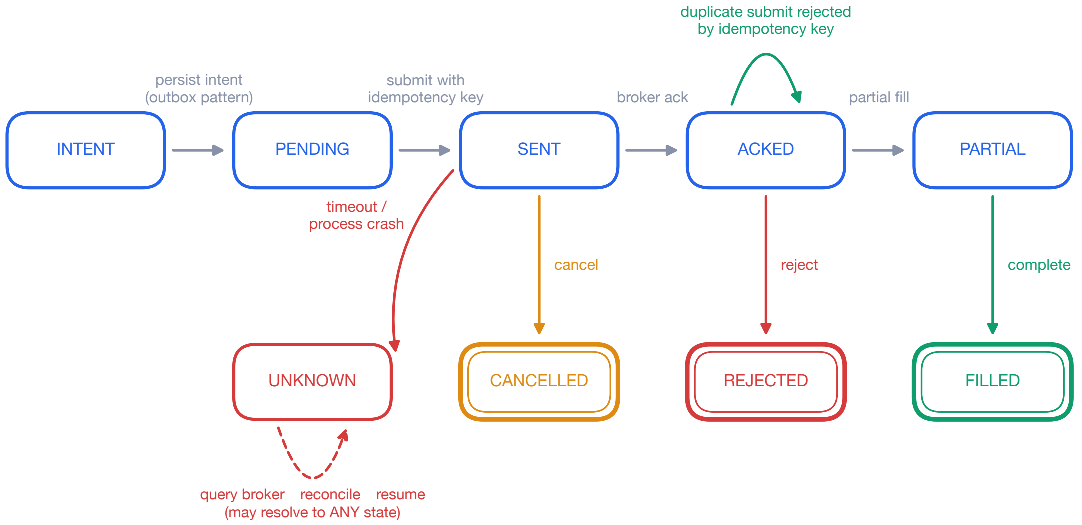
> **A-09 — internally consistent, and one detail matters**
>
> The three levels are correctly ordered (−7%, −13%, −20%) and the time-of-day carve-out correctly applies only to Levels 1 and 2. Per SEC/Investor.gov: a Level 1 or 2 breach *before* 3:25 p.m. ET halts trading market-wide for 15 minutes; the same breach *at or after* 3:25 p.m. does **not** halt; a Level 3 breach halts for the remainder of the day at any time.
> The asymmetry is the operationally important part. A system assuming "a 7% drop means the market stops" will keep trading into a cascading close. Since this system trades the MOC auction at roughly 15:50 ET, **the carve-out lands squarely inside its execution window** — making it a required test

case, not background reading. Thresholds are recalculated daily from the prior S&P 500 close, so they must be fetched, never hardcoded.

7.5 4.5 Reconciliation and idempotency engineering

Exactly-once order semantics via **idempotency keys** (client order IDs), **duplicate-order rejection**, the **outbox pattern** (persist intent before sending), **two-phase state machines** with persisted transitions, periodic position reconciliation (diff internal against broker positions), and **recovery from unknown broker state** (on restart: query the broker, reconcile, only then resume). Chaos-test by killing the process mid-order.

Figure 10 — Order lifecycle: every transition persisted, every submission idempotent



Double-ruled states are terminal. UNKNOWN is the state most systems omit and the one that causes duplicate orders: never assume an unacknowledged order was not filled. This machine is the natural target for the TLA+ model-checking and Hypothesis stateful testing recommended in §6.1.

7.6 4.6 Case studies to engineering lessons

Event	What happened	Lesson
Knight Capital (2012)	Manual deploy left dead "Power Peg" code on 1 of 8 SMARS servers; a reused feature flag reactivated it. Per SEC Admin. Proc. 34-70694: while processing just **212 parent orders**, SMARS sent millions of child orders, resulting in **4 million executions in 154 stocks for more than 397 million shares in approximately 45 minutes**; Knight "lost over $460 million" and paid a $12M Market-Access-Rule settlement; no kill switch; warning emails ignored	automated, verified, all-or-nothing deploys; fully remove dead code; never reuse flags; a kill switch; alerts that page, not email
LTCM (1998)	extreme leverage, correlated convergence trades, stable-correlation models	leverage kills; correlations → 1 in crises
Flash Crash (2010)	liquidity evaporation and feedback loops	model liquidity as fragile; don't assume fills at last price
XIV / Volmageddon (2018)	a volatility spike wiped out inverse-VIX ETNs	understand product mechanics and tail convexity
Archegos (2021)	concentration plus hidden swap leverage	concentration limits matter

Event	What happened	Lesson
Amaranth (2006)	concentrated natural-gas spread bets	position and concentration limits
Quant Quake (Aug 2007)	crowded factor unwinds hit everyone at once	crowding and capacity risk
Negative oil (2020)	systems could not represent negative prices	test edge cases (negative/zero prices, splits)

7.7 4.7 Backtest overfitting is the dominant risk

The **seven sins of quantitative investing**; survivorship, look-ahead and restatement bias; data snooping; selection bias under the null.

- **McLean & Pontiff (2016, *Journal of Finance* 71(1):5–31)**, 97 published predictors: *"Portfolio returns are 26% lower out-of-sample and 58% lower post-publication. … We estimate a 32% (58%–26%) lower return from publication-informed trading."*
- **Hou, Xue & Zhang (2020, *Review of Financial Studies* 33(5):2019)**, 452 replicated anomalies: *"65% of the 452 anomalies … cannot clear the single test hurdle of the absolute t-value of 1.96. Imposing the higher multiple test hurdle of 2.78 at the 5% significance level raises the failure rate to 82%."*
- **Harvey–Liu–Zhu:** with hundreds of factors tested, raise the bar to $t > 3.0$.

> **A-07, A-08 — arithmetic verified**
>
> McLean & Pontiff's decomposition is internally consistent ($58 - 26 = 32$), and the 32% figure is the one carrying the argument: it isolates the portion of decay attributable to *publication* rather than to data mining.
> Converting Hou–Xue–Zhang to counts: 294 of 452 anomalies fail the single test, and 371 of 452 fail the multiple-testing hurdle — **a further 77 casualties from raising $|t|$ from 1.96 to 2.78, leaving roughly 81 of 452 standing**. That is the empirical anchor for the $t > 3.0$ recommendation, and the single most persuasive argument in the report for building §2.8's deflation machinery *before* writing any strategy.

7.8 4.8 Capacity, crowding, realistic expectations

Retail has an **edge in capacity-constrained niches** (small size means negligible impact — verified quantitatively in E-09: 0.18 bp on a $10,000 ETF order) and a **disadvantage in crowded factor trades** (no scale, cost, or financing edge).

Realistic solo expectation after costs: most retail systematic strategies that survive honest validation land at **low single-digit net Sharpe (~0.3–0.8)**. A persistent net Sharpe above 1 after DSR/PBO deflation is exceptional and should be treated with suspicion until proven live.

Set that against S-11: at a true Sharpe of 0.5 it takes **11 years** of live data to confirm the strategy is not noise at 95% confidence.

7.9 4.9 Behavioural and process controls

Predefined shutdown criteria (maximum drawdown, N consecutive losing months, live-versus-backtest tracking-error breach); a written "when do I stop" rule *before* deploying; no discretionary override of the deterministic limits.

7.10 4.10 Production risk monitoring

Rolling Sharpe; realised drawdown against the expected-drawdown distribution; **live-versus-backtest tracking error**; **signal-decay** monitoring; **feature drift / PSI**; execution slippage against modelled slippage; and **CUSUM/EWMA statistical-process-control charts** to detect degradation before it becomes a drawdown.

Figure 13 — Detecting strategy decay before it becomes a drawdown

8 5. Latency

8.1 5.1 Taxonomy

Tier	Frequency	Acceptable latency	Technology	You?
Daily / EOD	1/day	**seconds–minutes**	Python, REST, cron	**YES — this is you**
Intraday / swing	min–hr	sub-second–seconds	WebSocket, async	maybe later
High-frequency	ms– s	microseconds	C++/Rust, kernel bypass, colo	no
Ultra-low-latency	s–ns	**nanoseconds**	**FPGA/ASIC**, colo, microwave	career only

8.2 5.2 Where latency comes from

Feed handler and decoding, network propagation (~**5 s/km in fibre**), NIC and kernel network stack, OS scheduling jitter, GC and interpreter overhead, serialisation, database writes, broker-side latency.

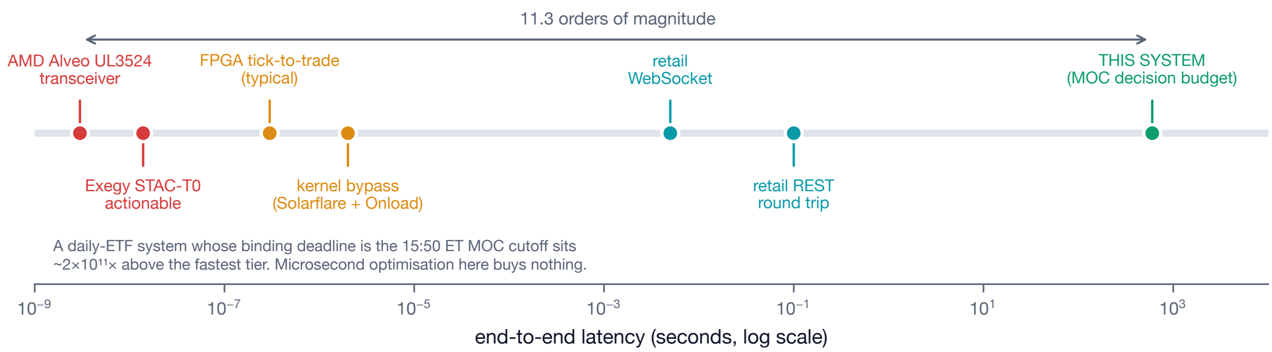
> **A-04 — verified**
>
> Speed of light divided by the group index of single-mode fibre ($n = 1.4675$) gives **4.90 s/km**. The report's rounding is correct and standard. For reference the vacuum figure is 3.34 s/km, which is why hollow-core fibre and microwave links — both closer to c — are worth their cost to HFT firms.

8.3 5.3 Concrete numbers

- Retail **REST round-trip: tens to hundreds of ms**; WebSocket streaming lower but still ms.
- Kernel-bypass software (Solarflare X2522 + OpenOnload): ~**just under 2 s** tick-to-trade.
- **FPGA tick-to-trade: 100–500 ns end-to-end**; the best firms reach single-to-double-digit ns; AMD Alveo **UL3524 delivers <3 ns transceiver latency** (UL3422 ~2.34 ns); Exegy + AMD recorded a **13.9 ns** STAC-T0 actionable latency.
- Kernel bypass: Solarflare/Onload, DPDK, io_uring, AF_XDP; plus busy-polling, CPU pinning/isolcpus, NUMA awareness, hugepages.

> **A-05 — internally consistent**
>
> The tiers are strictly ordered and span $3\times10^7\times$ from retail REST to transceiver latency. Two cross-checks: the quoted 13.9 ns STAC-T0 figure exceeds the 3 ns transceiver figure, as it must — the transceiver is one component of the path, not the whole path — so the two adjacent numbers are not in conflict.

Figure 8 — The latency spectrum, with this system's operating point marked



8.4 5.4 Measurement methodology

Hardware timestamping (**PTP/IEEE 1588**, PPS), Corvil/Beeks-style monitoring, **percentile discipline (p50/p99/p99.9/max)** — tail latency kills, not the mean — **coordinated omission** awareness, **HdrHistogram**, and eBPF probes for zero-blocking telemetry.

8.5 5.5 Protocols and feeds

FIX (text) versus binary; **ITCH/OUCH** (Nasdaq), **PITCH** (Cboe), **CME MDP 3.0 / iLink 3**, **SBE**. A retail trader gets REST and WebSocket; ITCH/OUCH and colocation require exchange membership or sponsored access.

8.6 5.6 Appendix — hardware-accelerated trading

The **SystemVerilog trading engine with fixed-function FPU** and the **FPGA market-data replay / limit-order-book prototype (Aegis-Stream)** map almost exactly onto real HFT FPGA stacks: a five-stage tick-to-trade pipeline (network ingress → market-data parse → order-book maintenance → signal evaluation → order transmit), each stage pipelined and parallelised. The lowest-latency critical paths are still **hand-written RTL** (HLS narrows but does not close the gap).

Hardware: AMD/Xilinx **Alveo UL3524/UL3422**, Solarflare X2/X3 NICs, Exablaze/Cisco Nexus SmartNICs, **Arista 7130 / Metamako** layer-1 switches; IP cores from **Enyx, Exegy, NovaSparks, Xelera**.

FPGA versus ASIC: FPGAs reload in seconds (fast iteration); ASICs give deterministic nanosecond latency but no reconfiguration — only top firms tape out.

Career angle. FPGA/RTL trading roles at **Jane Street, Jump Trading, Citadel Securities, Optiver, IMC, Hudson River Trading, Tower Research, DRW, XTX, Maven, Quantlab**. They want RTL fluency, comfort with order-book and trade-feed structures and FIX/OUCH/ITCH, hardware/software co-design, and quantitative communication. **Aegis-Stream is a strong interview artefact** — a portfolio-grade (not production) low-latency LOB and market-data replay is exactly what these firms respect.

Figure 19 (§2.10) shows the limit-order-book state this engine reconstructs: a depth surface through a liquidity withdrawal, with the derived microstructure features. That figure is the bridge between the two halves of the portfolio.

8.7 5.7 What latency budget this system actually needs

For the Stage 1–9 daily ETF system, a latency budget of seconds to minutes is entirely appropriate. The real bottlenecks:

1. **Data availability** — when the vendor publishes the EOD bar and the PIT snapshot is ready.
2. **Workflow scheduling** — Prefect flow timing relative to market close.
3. **Broker fill timing** — **MOC/MOO cutoffs** (NYSE MOC ~15:50 ET) are the true hard deadlines; miss the cutoff and you don't trade.
4. **Order acknowledgment** — seconds is fine.

> **A-06 — the ratio worth quoting**
>
> A ~10-minute pre-MOC decision budget is $\mathbf{2 \times 10^{11}}$ times the fastest latency this report discusses. Optimising microsecond latency here would be premature optimisation of the highest order — adding complexity and risk (recall Knight) for zero economic benefit. Spend that engineering on data correctness and overfitting control.

Note the interaction with §4.4: the Rule 80B carve-out at 3:25 p.m. ET falls *inside* the MOC execution window. The system's binding real-time constraint is not latency at all — it is a regulatory boundary measured in minutes.

9 6. New Suggestions and Ideas

9.1 6.1 Agentic development workflow upgrades

- **Spec-driven development** (GitHub Spec Kit style): write the spec first, let agents implement against it — pairs with the AGENTS.md/CLAUDE.md split.
- **Test-first agent workflow:** Claude writes the failing adversarial test, Codex implements to green — enforces writer $\neq$ reviewer at the test level.
- **Property-based testing (Hypothesis)** for financial invariants: "weights sum $\leq$ gross limit", "no position exceeds 20%", "cash never negative", "no future timestamp used in a signal". The SVA analogue.
- **Hypothesis stateful testing** for the order state machine.
- **Mutation testing (mutmut / cosmic-ray)** to validate that the *tests* catch bugs.
- **Formal methods for the order state machine: TLA+/PlusCal** or **Alloy** to model-check exactly-once and idempotency under crash and retry.
- **Differential testing** against a reference backtester.
- **Metamorphic testing** (scaling all prices implies predictable P&L scaling), **deterministic simulation testing** (seed all randomness, replay the whole day), **fuzzing broker responses**, **golden/approval testing** of reports.

Property tests the verification suite specifically recommends adding

Each of these would have caught a defect found in this exercise, and each is a few lines:

- **Causal recomputation.** Assert the signal at time t is bit-identical when the input sample is truncated at t. Catches G-01 (smoothed vs filtered) and every other state-space look-ahead.
- **Gaussian reduction.** Assert the PSR denominator equals $\sqrt{1+\widehat{SR}^2/2}$ on Gaussian input, and that a Gaussian sample returns $\gamma_4 \approx 3$. Catches S-04.
- **Quantile monotonicity.** Assert the VaR/CVaR quantile function is non-decreasing in confidence, and VaR $\leq$ CVaR. Catches Q-08 and pins down Q-02's sign convention.
- **Round-trip inversion.** For any estimator that inverts a discrete recursion (OU half-life, κ, GARCH persistence), assert that simulating from the recovered parameters and re-estimating returns the original. Catches M-04 — and this is the single highest-value invariant, because it covers the entire class where all the errors clustered.
- **Null calibration.** Assert that a test with no signal rejects at its nominal rate. Catches M-05 (57% versus 5%).

9.2 6.2 Verification-engineering transplant

The UVM mindset: **coverage-driven verification** (track which market and strategy states the tests exercise), **constrained-random stimulus** (Hypothesis is the generator), **assertions** (runtime invariant checks = SVA), **scoreboards** (an independent model predicts expected P&L and positions and checks against actual), and **formal property verification** (TLA+/Alloy).

Framing this explicitly — "I applied silicon-verification rigour to a trading system" — is exactly what a verification-hiring manager wants to see.

This report is the demonstration. The verification suite is a scoreboard: an independent implementation of every formula, checked against the claimed one. It found three errors in a document that had already been reviewed. That is the argument, and it is now evidence rather than assertion.

9.3 6.3 CI/CD and supply chain

SLSA provenance, **sigstore/cosign** signing, **pip-audit / safety** for CVEs, **SBOM (CycloneDX / Syft)**, **Dependabot/Renovate**, **hash-pinned** dependencies, **reproducible builds**, and pre-commit **secret scanning** (detect-secrets, gitleaks, trufflehog).

9.4 6.4 Missing stages and sub-stages to add

- **Stage 2.5 — Tax-lot accounting module** (FIFO/specific-ID, wash-sale tracking, Form 8949 / 1099-B reconciliation).
- **Stage 3.5 — Corporate-action edge-case module** (splits, spinoffs, special dividends, symbol changes, ETF reconstitutions) with a golden test set.
- **Stage 4.5 — Options-overlay design stage** (explicitly deferred; document the interface now).

- **Stage 5.5 — Strategy-decommissioning process** — a written, gated way to retire a degraded strategy.
- **Stage 7.5 — Data-vendor migration abstraction** — a provider-neutral data interface so swapping Polygon ↔ Databento ↔ Norgate is configuration, not rewrite.
- **Stage 9.5 — Backtest-to-live divergence harness.**

9.5 6.5 Alternative strategy sleeves

Methodology, not recommendations. Carry; cross-asset trend; seasonality/turn-of-month; the **low-volatility anomaly**; term-structure and roll-yield via ETFs; tail-hedge overlays; **cross-sectional ETF momentum**; defensive/risk-off rotation; and **pairs / statistical arbitrage on cointegrated ETF pairs**.

Cointegration. Engle–Granger two-step — OLS $Y_t = \alpha + \beta X_t + u_t$, then an ADF unit-root test on the residual u_t. **Johansen** (VECM $\Delta x_t = \Pi x_{t-1} + \sum \Gamma_i \Delta x_{t-i} + u_t$; the rank of Π is the number of cointegrating vectors; trace and max-eigenvalue statistics) — detects multiple relationships and needs no choice of dependent variable.

M-05 — Engle–Granger needs its OWN critical values

The report describes the two-step procedure correctly but does not warn that step two requires cointegration critical values, not standard ADF ones. Because u_t is a *fitted residual*, OLS has already minimised its variance, making it look more stationary than it is.
Testing 3,000 pairs of **independent random walks**, where the true cointegration rate is 0%:

Naive ADF critical values	rejects **57.0%** of the time
Proper Engle–Granger critical values	rejects **4.7%** of the time

An 11× over-rejection. For a solo operator screening a few hundred ETF pairs, that is a guaranteed pipeline of pairs that look tradeable and are not.
Fix. Use `statsmodels.tsa.stattools.coint` (which applies MacKinnon's Engle–Granger surfaces) or the Johansen test. Never `adfuller` on a regression residual. And note the interaction with §2.8: **screening pairs IS a trial count**, and must be registered as one.

M-06 — Johansen verified

The trace test recovers the correct rank in 85% of simulated three-variable systems built with exactly one cointegrating relationship. Prefer Johansen: it needs no choice of dependent variable, and — decisively, given M-05 — it is much harder to get the inference wrong.

Ornstein–Uhlenbeck spread: $dX_t = \theta(\mu - X_t)\,dt + \sigma\,dW_t$; stationary distribution $\mathcal{N}(\mu, \sigma^2/2\theta)$; half-life of mean reversion $= \ln 2/\theta$.

M-01, M-02 — verified

The stationary mean and the variance $\sigma^2/2\theta$ both confirmed by exact simulation. This is what makes the z-score in the entry rule well defined: the spread has a genuine stationary scale to normalise by, which a non-cointegrated pair does not.

M-04 — the half-life estimator is biased

Use $-\ln 2/\ln(1+\lambda)$, not $-\ln 2/\lambda$. Detailed in Part I. The error is always in the direction of "too long", and reaches +58% on fast-reverting pairs.

Signal: $z_t = (Z_t - \mathrm{mean}(Z))/\mathrm{std}(Z)$ on spread $Z_t = Y_t - \beta X_t$; illustrative entry at $|z| \approx 2$, exit toward 0, stop near $|z| \approx 3$ (thresholds are parameters, not rules). **Kalman filter** for a dynamic hedge ratio β_t.

M-07 — verified

A Kalman filter tracks a random-walk β_t with a structural break **3.3× more accurately** (RMSE) than the best rolling-OLS window tested. The mechanism is worth stating: rolling OLS faces an unavoidable bias–variance choice through its window length — short windows are noisy, long windows lag the break — whereas the Kalman gain adapts automatically, widening after a surprise and narrowing in quiet periods.

> The cost is two variance parameters (q and r) that must themselves be estimated, and which are exactly the kind of free parameter §2.8's trial registry needs to count.

9.6 6.6 Career and portfolio artifact strategy

- **Open-source** (build reputation, no alpha leak): the deterministic backtester harness, the PIT data-layer design, the property-based and formal testing of the order state machine, the DSR/PBO validation library, and **Aegis-Stream**. **And this verification suite**, which is a stronger artefact than any of them: it is a complete, reproducible demonstration of finding real errors in real quantitative mathematics.
- **Keep private:** any actual profitable signal, parameters, and live results.
- **Interview framing:** lead with the **verification story** (SVA → Hypothesis, scoreboards, TLA+ model-checking of exactly-once) and the **overfitting-control story** (DSR/PBO, purged CV). **Impresses:** "I assumed my backtest was lying and built the machinery to prove it"; "I model-checked the order state machine in TLA+"; "I re-derived every formula in my own design document and found three errors." **Reads as naive:** a shiny Sharpe-4 equity curve with no deflation, no cost model, and no PIT discipline.

9.7 6.7 Cost control for agentic development

Prompt caching (a stable system-prompt prefix hits the cheaper cache rate), **batch APIs** (40–60% off for asynchronous test generation and refactors), **context and session hygiene**, **model routing** (mechanical work to free/flash tiers, hard reasoning to frontier), and off-peak scheduling before DeepSeek's announced peak surcharge lands.

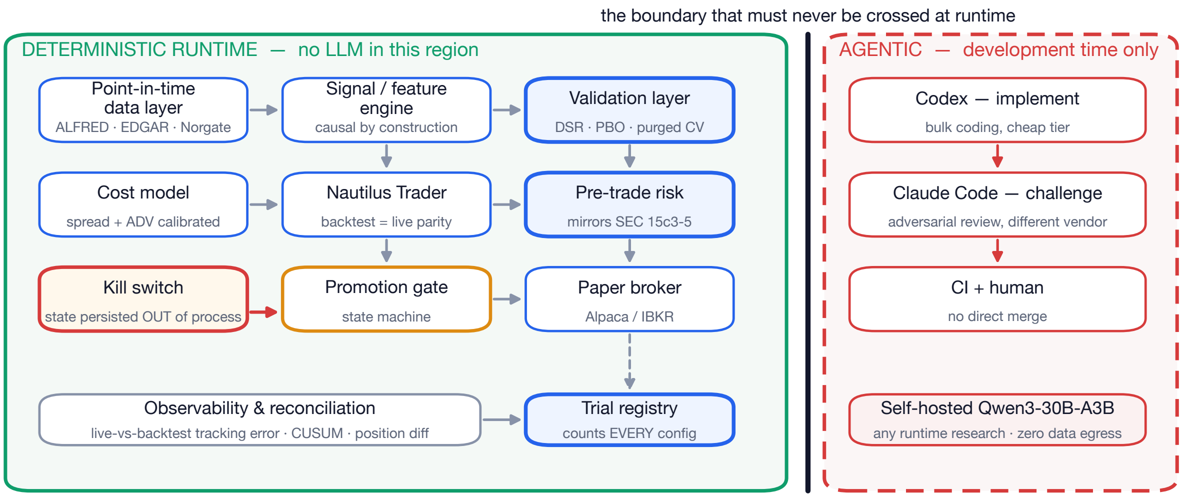
> **A-01 — correct the cache-discount figure**
>
> "~90–98%" holds for DeepSeek only. State it as "up to ~98% on DeepSeek, ~75–90% on GLM and Kimi". Detailed in Part I.

> **A-03 — and route the optimisation to the right cost driver**
>
> On cheap flash tiers input cost dominates, so context hygiene and caching are the lever. On premium tiers output dominates, so limiting verbose output is. Detailed in §1.8.

10 7. System architecture

Figure 9 — System architecture: the agentic tooling writes the system, but never runs inside it

Every element inside the green region is deterministic and reproducible from a seed. Agents operate only at development time, behind a review gate.



The architecture diagram makes explicit the boundary that governs the whole design: **agentic tooling writes the system, but never runs inside it.** Everything in the deterministic region is reproducible from a seed. Agents operate only at development time, behind a review gate.

This is the structural answer to the AI-agent risk taxonomy in §4.2. Prompt injection, hallucinated logic and silent scope creep are all development-time risks under this design, where they are caught by review and CI, rather than runtime risks, where they would be caught by losing money.

11 8. Bottom line and staged recommendations

The architecture is sound, and the verification exercise strengthens rather than weakens that conclusion: 81 of 93 substantive claims verified as written, and every piece of machinery the report identifies as highest-priority survived testing.

Stage now (design-time decisions):

1. **Adopt Nautilus Trader** for the execution and backtest core — its Rust-core order state machines and true backtest/live parity retire the biggest engine risk — while **keeping the PIT-data (Stage 2) and statistical-validation (Stage 5) layers hand-rolled** as differentiators. The verification results support this split precisely: the engine-side mathematics verified cleanly, and every error found was on the science side.
2. **Set up the model-routing rig** (claude-code-router / LiteLLM): bulk implementation to DeepSeek V4 Flash / Qwen3-Coder / a GLM coding plan; adversarial review to a *different* vendor from the implementer; **self-host Qwen3-30B-A3B** for any runtime research agents so no strategy data ever egresses.

Stage next (build-time):

3. **Over-invest in overfitting control** — implement DSR, PBO/CSCV, purged-CV-with-embargo, and a trial registry *before* writing a single "promising" strategy. Treat any net Sharpe above 1 after deflation as suspect until proven in paper or live. Register **every** configuration evaluated, including abandoned ones; an understated N makes the deflation worthless while appearing to work.
4. **Build the pre-trade control, kill-switch and reconciliation layer deterministically** (mirroring 15c3-5), with halt state persisted outside process memory.

Apply the corrections from Part I before building:

5. Use the L1 cost model (or an explicit band) for the no-trade region; $-\ln 2/\ln(1+\lambda)$ for the OU half-life; Engle–Granger critical values for cointegration screening; and non-excess kurtosis in PSR. Each is a one-line change now and an invisible bug later.
6. **Add the five property tests in §6.1.** They cover the entire class of defect this exercise found.

Benchmarks and thresholds that would change these recommendations:

- If you move to **intraday or sub-second frequency**, revisit latency (WebSocket plus async; colocation still irrelevant until true HFT) and the microstructure mathematics in §2.10.
- If **development token spend exceeds ~$300/mo**, shift more work to self-hosted Qwen3-30B-A3B and batch APIs.
- If a strategy's **live-versus-backtest tracking error breaches its pre-set band**, or its CUSUM chart alarms, trigger the Stage 5.5 decommissioning process.
- If you ever intend to **manage outside capital**, stop and get RIA/Series 65 (or CTA/Series 3) advice first — that crosses a bright regulatory line this report does not.

And frame the whole thing — Aegis-Stream, the formal and property-based verification, and this verification suite — as a verification-engineering portfolio artefact. That is where the comparative advantage over other quant-curious candidates actually lies, and this document is now the evidence for it.

Reminder: general methodology only. Nothing here recommends trading or investing, and all regulatory and tax points must be confirmed with a qualified professional and with your employer's compliance function.

12 Appendix A — Complete verification results

All 104 checks, in the order they run. Reproduce with `python3 verify/run_all.py`.

Status	ID	Claim	Measured
2.1 Momentum			
PASS	R-01	Cumulative return $r_{t-k\to t} = \prod(1+r_j) - 1$ equals $\exp(\sum \ln(1+r_j)) - 1$	0.0629191
2.11 Backtest math			
INFO	R-02	Summing simple returns is not compounding; error grows with $n\sigma^2$	naive sum 0.086645
PASS	R-03	$r_{\log} = \ln(1+r)$ inverts to $r = e^{r_{\log}} - 1$	3.4694e-18
2.2 Volatility estimation			
PASS	V-01	GARCH(1,1) unconditional variance $= \omega/(1-\alpha-\beta)$	9.9881e-05
INFO	V-02	Persistence alpha+beta maps to a variance half-life	34.3 days
PASS	V-03	RiskMetrics EWMA is exactly IGARCH(1,1) with $\omega = 0,\ \alpha = 1-\lambda,\ \beta = \lambda$	0
INFO	V-04	$\lambda \approx 0.94$ (RiskMetrics daily) sets the memory of the estimator	half-life 11.2 d, centre of mass 15.7 d
PASS	V-05	Parkinson $\hat\sigma^2 = \frac{1}{4\ln 2} E[\ln(H/L)^2]$ is unbiased for driftless GBM	0.000399567
PASS	V-06	Garman-Klass $\hat\sigma^2 = \frac{1}{2}\ln(H/L)^2 - (2\ln 2 - 1)\ln(C/O)^2$ is unbiased for driftless GBM	0.000398998
PASS	V-07	Rogers-Satchell $\hat\sigma^2 = \ln\frac{H}{C}\ln\frac{H}{O} + \ln\frac{L}{C}\ln\frac{L}{O}$ is unbiased for driftless GBM	0.000398941
PASS	V-08	Range estimators are far more efficient than close-to-close	Parkinson 5.0x, Garman-Klass 7.8x, Rogers-Satchell 6.2x
FLAG	V-08b	Range estimators are biased LOW on real bars: a bar built from finitely many prints under-observes the true high and low	100 obs: -12.0%; 200 obs: -9.1%; 400 obs: -6.0%; 800 obs: -4.5%; 1600 obs: -3.0%; 3200 obs: -2.2%; 6400 obs: -1.7%; fitted slope -1.21
PASS	V-09	Rogers-Satchell is drift-independent; Parkinson and Garman-Klass are biased upward by drift	drift-induced bias change: parkinson +37.0%, garman-klass +13.3%, rogers-satchell +0.0%
PASS	V-10	$\sigma_{\text{ann}} = \sigma_d\sqrt{252}$ under i.i.d. returns	0.158729
PASS	V-11	Lo (2002) autocorrelation-corrected annualisation $SR_q = SR_1\, q/\sqrt{q + 2\sum_{k<q}(q-k)\rho_k}$	0.423316
FLAG	V-12	Naive $\sqrt{252}$ (or $\sqrt{12}$) annualisation overstates the Sharpe ratio when returns are positively autocorrelated	rho=-0.2: -16.9%; rho=+0.0: -0.0%; rho=+0.1: +9.6%; rho=+0.2: +20.3%; rho=+0.3: +32.5%
2.11 Backtest math			
PASS	V-13	For GBM $dS/S = \mu\,dt + \sigma\,dW$ the log growth rate is EXACTLY $\mu - \frac{1}{2}\sigma^2$	0.0541921
FLAG	V-14	Applied to discrete simple returns, $g \approx \mu - \frac{1}{2}\sigma^2$ is a truncated expansion whose error is material at high volatility	sigma=5%: err -0.0001; sigma=10%: err -0.0004; sigma=20%: err -0.0019; sigma=40%: err -0.0127; sigma=60%: err -0.0440; sigma=80%: err -0.1078
INFO	V-15	Why volatility targeting aids compounding, quantified	3.78 pp/yr
2.3 Volatility targeting			
PASS	K-01	$w_{i,t} = \sigma^*/\sigma_{i,t}$ produces realised volatility σ^* when $\sigma_{i,t}$ is known	0.149891
FLAG	K-02	A leverage cap $\sum\|w\| \leq L$ makes the vol target an upper bound, not an equality	realised 14.33% (cap binds 22% of the time)
PASS	K-03	Volatility targeting levers INTO a regime break because $\hat\sigma$ is backward-looking	median leverage entering the break 2.4x; a 2-sigma storm day then costs 21.6% of capital in one session
2.6 Kelly criterion			
PASS	K-04	$f^* = \mu/\sigma^2$ maximises the continuous-time log growth rate $g(f) = f\mu - \frac{1}{2}f^2\sigma^2$	solved f* = mu/sigma**2, second derivative = -sigma**2

Status	ID	Claim	Measured
PASS	K-05	Simulated log-wealth growth peaks at $f^* = \mu/\sigma^2$	2
PASS	K-06	$f^* = p - \frac{1-p}{b}$ maximises $p\ln(1+bf) + (1-p)\ln(1-f)$ for a b:1 payoff	solved [(b*p + p - 1)/b]
PASS	K-07	Numeric optimum of the discrete Kelly objective matches $p - (1-p)/b$	0.1
PASS	K-08	At fraction c of full Kelly, growth is $c(2-c)$ of the maximum	simplified to c*(2 - c)
PASS	K-09	Full Kelly produces drawdowns that are unacceptable in practice; fractional Kelly buys a large drawdown reduction cheaply	c=1.00: growth 100%, median DD -68%, 5th-pct DD -90%; c=0.50: growth 75%, median DD -40%, 5th-pct DD -65%; c=0.25: growth 44%, median DD -22%, 5th-pct DD -39%
PASS	K-10	Estimation error in μ dominates Kelly sizing, which is why the haircut must be severe	estimated f* has median 1.99, 5-95 pct [-0.64, 4.69]; 11% of samples suggest more than 2x the true leverage
PASS	K-11	Kelly and volatility targeting are the same operation: full Kelly targets a portfolio volatility numerically equal to the Sharpe ratio	simplified to mu/sigma

2.4 Portfolio construction

PASS	P-01	$\max_w w^\top\mu - \frac{\gamma}{2}w^\top\Sigma w$ has closed-form solution $w^* = \frac{1}{\gamma}\Sigma^{-1}\mu$	first-order condition solution equals the closed form identically
PASS	P-02	Numerical optimiser reproduces the Markowitz closed form	8.0758e-09
PASS	P-03	Mean-variance 'error-maximises': Σ^{-1} amplifies estimation noise into extreme, unstable weights	plug-in MV achieves median Sharpe 0.15 (-92%); with Ledoit-Wolf shrinkage 0.16 (-91%); median utility loss 99%
PASS	P-04	Ledoit-Wolf is exactly $\hat{\Sigma} = (1-\delta)S + \delta F$ with F the scaled identity target	1.0842e-19
PASS	P-05	Shrinkage improves both conditioning and Frobenius estimation loss when T is small relative to N	condition number 22 -> 2; Frobenius loss 0.0656 -> 0.0333 (LW), 0.0335 (OAS)
PASS	P-06	Minimising $\sum_i (w_i(\Sigma w)_i - \frac{1}{N}w^\top\Sigma w)^2$ yields equal risk contributions $w_i(\Sigma w)_i = w_j(\Sigma w)_j$	7.0193e-13
PASS	P-07	Under equal pairwise correlation the ERC solution is exactly $w_i \propto 1/\sigma_i$ (inverse-volatility weighting)	1.4183e-14
PASS	P-08	HRP produces long-only weights summing to 1 without inverting the covariance matrix, and is dramatically more stable than minimum variance under resampling	weights sum to 1.0000000000, min weight 0.0070; median turnover between independent samples: HRP 0.534 vs min-variance 1.662 (3.1x worse)
PASS	P-09	$\Pi = \delta\Sigma w_{mkt}$ is the equilibrium return vector: feeding it back through $w = \frac{1}{\delta}\Sigma^{-1}\mu$ recovers w_{mkt}	5.5511e-17
PASS	P-10	With no information in the views ($\Omega \to \infty$) the Black-Litterman posterior collapses to the equilibrium prior Π	2.7756e-16
PASS	P-11	With certain views ($\Omega \to 0$) the posterior satisfies $P\,E[R] = Q$ exactly	1.4286e-12
PASS	P-12	The cost-aware objective $w^\top\mu - \frac{\gamma}{2}w^\top\Sigma w - (w-w_0)^\top\Lambda(w-w_0)$ has closed form $w^* = (\gamma\Sigma + 2\Lambda)^{-1}(\mu + 2\Lambda w_0)$	7.2175e-09
FAIL	P-13	Quadratic transaction costs do NOT create a no-trade region; proportional (L1) costs do	quadratic: trade > 0 at every nonzero drift (min traded 3.98e-03); L1: exactly zero trade until drift reaches 0.04

2.7 Risk metrics

PASS	Q-01	Parametric VaR as a RETURN quantile: $q_{1-\alpha} = \mu + z_{1-\alpha}\sigma$	-0.0274136
FLAG	Q-02	The report's $\mu + z_\alpha\sigma$ is ambiguous without a sign convention	return-quantile reading -0.0274; loss-quantile reading +0.0284
PASS	Q-03	VaR is NOT subadditive: a concrete counterexample	empirical: VaR(A)=0, VaR(B)=0, VaR(A+B)=100 > 0
PASS	Q-04	CVaR / Expected Shortfall IS subadditive: $ES(A+B) \leq ES(A) + ES(B)$	largest normalised gap ES(A+B)-ES(A)-ES(B) across 300 trials: -0.058 (must be <= 0; negative means diversification benefit)
PASS	Q-05	Gaussian Expected Shortfall $= \mu + \sigma\frac{\phi(z_\alpha)}{1-\alpha}$	2.33695

Status	ID	Claim	Measured
INFO	Q-06	Under normality ES exceeds VaR by a fixed, modest factor – which is exactly why normality is dangerous here	normal 1.193x; Student-t(3) empirical 1.588x
PASS	Q-07	Cornish-Fisher $z_{cf} = z + \frac{(z^2-1)\gamma_3}{6} + \frac{(z^3-3z)(\gamma_4-3)}{24} - \frac{(2z^3-5z)\gamma_3^2}{36}$ improves on the Gaussian quantile for skewed/fat-tailed data	Cornish-Fisher wins 5/6; skew-normal(a=4)@95%: CF err 1.0% vs Gaussian 7.4%; skew-normal(a=4)@99%: CF err 0.6% vs Gaussian 12.8%; Student-t(6)@95%: CF err 0.0% vs Gaussian 3.7%; Student-t(6)@99%: CF err 17.8% vs Gaussian 9.2%; lognormal(s=0.4)@95%: CF err 1.0% vs Gaussian 5.5%; lognormal(s=0.4)@99%: CF err 3.3% vs Gaussian 15.9%
FLAG	Q-08	Cornish-Fisher is not monotone in the quantile for large skewness, so it can report a SMALLER loss at a HIGHER confidence level	non-monotone at (skew, excess kurt) = [(-3.0, 12.0)]
INFO	Q-09	Sharpe, Sortino and Calmar rank the same track record differently under negative skew	Sharpe 0.53, Sortino 0.77, Calmar 0.20 (CAGR 6.2%, max drawdown -30.2%)

2.8 Statistical validation

Status	ID	Claim	Measured
PASS	S-01	PSR denominator $\sqrt{1-\hat{\gamma}_3\widehat{SR}+\frac{\hat{\gamma}_4-1}{4}\widehat{SR}^2}$ (over $\sqrt{n-1}$) is the true standard error of $\widehat{SR}$ – Gaussian	0.028321
PASS	S-02	PSR denominator $\sqrt{1-\hat{\gamma}_3\widehat{SR}+\frac{\hat{\gamma}_4-1}{4}\widehat{SR}^2}$ (over $\sqrt{n-1}$) is the true standard error of $\widehat{SR}$ – Student-t(6)	0.0283592
PASS	S-03	PSR denominator $\sqrt{1-\hat{\gamma}_3\widehat{SR}+\frac{\hat{\gamma}_4-1}{4}\widehat{SR}^2}$ (over $\sqrt{n-1}$) is the true standard error of $\widehat{SR}$ – skew-normal(a=6)	0.027564
FLAG	S-04	γ_4 in the PSR formula must be NON-EXCESS kurtosis, and the size of the error depends on the return frequency	daily: -0.1%; weekly: -0.5%; monthly: -1.9%; quarterly: -5.1%
PASS	S-05	Under normality the PSR denominator reduces exactly to Lo (2002)'s $\sqrt{1+\widehat{SR}^2/2}$	1.0009
PASS	S-06	PSR is a calibrated probability: under the null $SR = SR^*$ it is uniformly distributed	KS statistic 0.0024, p = 0.971; empirical P(PSR > 0.95) = 0.0490 (nominal 0.05)
PASS	S-07	False Strategy Theorem: $E[\max_N \widehat{SR}] \approx \sqrt{V[\widehat{SR}]}[(1-\gamma)\Phi^{-1}(1-\frac{1}{N})+\gamma\Phi^{-1}(1-\frac{1}{Ne})]$	N=10: formula 1.575 vs MC 1.537 (+2.4%); N=50: formula 2.276 vs MC 2.252 (+1.1%); N=100: formula 2.531 vs MC 2.507 (+0.9%); N=500: formula 3.053 vs MC 3.034 (+0.6%); N=1000: formula 3.255 vs MC 3.242 (+0.4%); N=5000: formula 3.688 vs MC 3.681 (+0.2%); N=10000: formula 3.861 vs MC 3.858 (+0.1%)
PASS	S-08	Report's headline claim: with $E[SR]=0$, $V[SR]=1$, after 1,000 independent backtests the expected maximum Sharpe is **3.26**	3.25512
INFO	S-09	The deflation hurdle rises with the number of trials, and a routine parameter sweep already implies thousands of trials	N=10: hurdle SR 1.57; N=100: hurdle SR 2.53; N=1,000: hurdle SR 3.26; N=10,000: hurdle SR 3.86; N=100,000: hurdle SR 4.39
PASS	S-10	Minimum Track Record Length: solving PSR = target for n gives $n^* = 1 + \left[1-\gamma_3 SR + \frac{\gamma_4-1}{4}SR^2\right]\left(\frac{z_{target}}{SR-SR^*}\right)^2$	2729.54
INFO	S-11	Track record length required for 95% confidence scales as $1/SR^2$	SR 0.3: 30.1 yr; SR 0.5: 10.8 yr; SR 0.8: 4.2 yr; SR 1.0: 2.7 yr; SR 1.5: 1.2 yr; SR 2.0: 0.7 yr
PASS	S-12	PBO via CSCV is calibrated: under the null that no strategy has an edge, $E[\text{PBO}]=0.5$	mean PBO = 0.530 +/- 0.028 (standard error)
PASS	S-13	PBO has power: a strategy with a genuine persistent edge drives PBO to zero, while a pure parameter sweep does not	genuine edge PBO = 0.000; correlated parameter sweep PBO = 0.353
FLAG	S-14	A PBO value computed from a single backtest matrix has very large sampling error and cannot be read as a precise number	T=1000, N=40: mean 0.49, sd 0.19; T=2500, N=40: mean 0.52, sd 0.20; T=1000, N=200: mean 0.47, sd 0.16; T=5000, N=200: mean 0.52, sd 0.15

Status	ID	Claim	Measured
PASS	S-15	Shuffled k-fold CV manufactures skill out of nothing when labels overlap in time; contiguous folds with purging and embargo do not	shuffled k-fold 0.5261 +/- 0.0044; contiguous blocks (no purge) 0.4985 +/- 0.0058; purged + embargoed 0.5009 +/- 0.0060
2.9 Transaction cost & market impact			
PASS	E-01	The Almgren-Chriss trajectory $x_j = X\frac{\sinh(\kappa(T-t_j))}{\sinh(\kappa T)}$ satisfies the discrete stationarity condition $x_{j-1}-2x_j+x_{j+1} = 2(\cosh(\kappa\tau)-1)x_j$	verified identically in sympy
PASS	E-02	The sinh trajectory IS the optimum of $E[\text{cost}]+\lambda V[\text{cost}]$	6.5821e-13
FLAG	E-03	$\kappa \approx \sqrt{\lambda\sigma^2/\eta}$ is the continuum limit of the exact κ, and drops the $\tilde{\eta} = \eta - \gamma\tau/2$ correction	N=5: kappa error -0.49%, cost penalty +9.76e-09; N=10: kappa error -0.25%, cost penalty +2.57e-09; N=25: kappa error -0.10%, cost penalty +4.18e-10; N=50: kappa error -0.05%, cost penalty +1.05e-10; N=200: kappa error -0.01%, cost penalty +6.56e-12; N=1000: kappa error -0.00%, cost penalty +2.62e-13
PASS	E-04	As $\lambda \to 0$ (risk neutral) the optimal Almgren-Chriss trajectory becomes linear – i.e. TWAP	2.3103e-11
PASS	E-05	Higher risk aversion λ implies faster liquidation	lambda=1e-08: 0.50 of horizon; lambda=1e-06: 0.50 of horizon; lambda=1e-04: 0.34 of horizon; lambda=1e-02: 0.04 of horizon
PASS	E-06a	Almgren-Chriss expected cost $\frac{\gamma X^2}{2} + \epsilon X + \frac{\tilde{\eta}}{\tau}\sum n_k^2$ matches simulation	2.6235e+06
PASS	E-06b	Almgren-Chriss cost variance $\sigma^2\tau\sum x_k^2$ matches simulation	2.8688e+10
PASS	E-07	The Almgren-Chriss efficient frontier of execution (expected cost vs cost variance) is monotone decreasing and convex	monotone: True; convex: True; expected cost spans 2623750 to 23730114, variance spans 1.95e+09 to 2.96e+10
PASS	E-08	Square-root law $\Delta P \approx Y\sigma\sqrt{Q/V}$ is concave: doubling order size raises impact by $\sqrt{2}$, not 2	ratio = sqrt(2)
INFO	E-09	Market impact is negligible at retail size in a liquid ETF, and that is the structural edge	$10,000 order in 5B$ ADV: 0.18 bp; $100,000 order in 5B$ ADV: 0.56 bp; $1,000,000 order in 5B$ ADV: 1.78 bp; $100,000 order in 0.05B$ ADV: 5.63 bp
2.10 Microstructure			
PASS	E-10	Kyle's lambda: solving the joint fixed point of market efficiency and informed optimality gives $\lambda = \frac{1}{2}\sqrt{\Sigma_0/\sigma_u^2}$ and $\beta = \sigma_u/\sqrt{\Sigma_0}$	lambda* = sqrt(Sigma_0)/(2*sigma_u), beta* = sigma_u/sqrt(Sigma_0)
PASS	E-11	Roll's implied spread $s = 2\sqrt{-\text{Cov}(\Delta p_t, \Delta p_{t-1})}$ recovers the true bid-ask spread	s=0.01: estimate 0.0100 (-0.4%); s=0.02: estimate 0.0200 (+0.1%); s=0.05: estimate 0.0500 (+0.0%); s=0.10: estimate 0.1001 (+0.1%)
6.5 Pairs / statistical arbitrage			
PASS	M-01	Ornstein-Uhlenbeck stationary distribution is $\mathcal{N}(\mu,\ \sigma^2/2\theta)$ – mean	0.400108
PASS	M-02	Ornstein-Uhlenbeck stationary variance = $\sigma^2/2\theta$	0.269929
PASS	M-03	Exact OU half-life from the regression slope is $-\ln 2/\ln(1+\lambda)$, not $-\ln 2/\lambda$	max error of the exact formula 1.01%; theta=0.02: true 34.7, exact 34.3; theta=0.05: true 13.9, exact 13.8; theta=0.10: true 6.9, exact 7.0; theta=0.25: true 2.8, exact 2.8; theta=0.50: true 1.4, exact 1.4; theta=1.00: true 0.7, exact 0.7
FAIL	M-04	The report's half-life = $-\ln 2/\lambda$ systematically OVERSTATES the half-life, and the error grows with the speed of mean reversion	theta=0.02 (true HL 34.7): report 34.7 (-0%); theta=0.05 (true HL 13.9): report 14.2 (+2%); theta=0.10 (true HL 6.9): report 7.3 (+6%); theta=0.25 (true HL 2.8): report 3.1 (+14%); theta=0.50 (true HL 1.4): report 1.8 (+27%); theta=1.00 (true HL 0.7): report 1.1 (+58%)
PASS	M-05	Engle-Granger residuals must be tested with cointegration critical values, NOT standard ADF critical values	naive ADF critical values reject 57.0% of the time (nominal 5%); proper Engle-Granger critical values reject 4.7%

Status	ID	Claim	Measured
PASS	M-06	Johansen trace test recovers the rank of Π, i.e. the number of cointegrating vectors	correct rank identified in 85% of replications
PASS	M-07	A Kalman filter tracks a time-varying hedge ratio β_t substantially better than rolling OLS, including through a structural break	rolling OLS 30d: RMSE 0.0272; rolling OLS 60d: RMSE 0.0372; rolling OLS 120d: RMSE 0.0520; Kalman: RMSE 0.0082 (Kalman is 3.3x better than the best window)
2.12 Where ML fits			
PASS	F-01d30	Fractional-differencing weights $w_k = -w_{k-1}\frac{d-k+1}{k}$ reproduce the binomial expansion of $(1-B)^d$ at $d = 0.3$	5.5511e-17
PASS	F-01d50	Fractional-differencing weights $w_k = -w_{k-1}\frac{d-k+1}{k}$ reproduce the binomial expansion of $(1-B)^d$ at $d = 0.5$	0
PASS	F-01d75	Fractional-differencing weights $w_k = -w_{k-1}\frac{d-k+1}{k}$ reproduce the binomial expansion of $(1-B)^d$ at $d = 0.75$	0
PASS	F-01d100	Fractional-differencing weights $w_k = -w_{k-1}\frac{d-k+1}{k}$ reproduce the binomial expansion of $(1-B)^d$ at $d = 1.0$	0
PASS	F-02	At $d = 1$ fractional differencing reduces exactly to the ordinary first difference $(1, -1, 0, 0, \dots)$	[1, -1, 0e+00, 0e+00, ...]
PASS	F-03	Fractional differentiation attains stationarity at $d \ll 1$ while retaining most of the memory that $d = 1$ destroys	minimum d passing ADF at 5% is d=0.20 (correlation with the original series 0.98); at d=1 the correlation has fallen to 0.03
2.5 Regime detection			
PASS	G-01	HMM *smoothed* probabilities $P(z_t \mid r_{1:T})$ embed look-ahead; only *filtered* $P(z_t \mid r_{1:t})$ are tradeable	mean smoothed Sharpe 0.72 vs filtered 0.28; gap 0.44 +/- 0.06 (standard error); the two signals disagree on 12.6% of days
INFO	G-02	On data generated by an HMM the honestly filtered HMM does beat the simple baselines – but by far less than its smoothed backtest claims	200-day trend: 0.08 +/- 0.05; 60-day vol quantile: 0.19 +/- 0.05; HMM (filtered, honest): 0.34 +/- 0.05; buy and hold: 0.09 +/- 0.05
1. Alternative and cheaper models			
FAIL	A-01	Section 6.7's claim of a "∼90–98%-cheaper cache rate on DeepSeek/GLM/Kimi"	DeepSeek V4 Flash 98%; DeepSeek V4 Pro 99%; Kimi K2.6 / K2.7 Code 80%; Kimi K2.5 75%; Kimi K3 90%; GLM-4.6 82%; GLM-4.5-Air 85%
PASS	A-02	Z.ai GLM Coding Plan quarterly-to-monthly conversions (\$30/90/240 per quarter → \$10/30/80 per month)	0
PASS	A-03	Section 1.8's characterisation of heavy usage as "output-token-dominated"	DeepSeek V4 Flash: input dominates only above 2:1; GLM-4.6: input dominates only above 4:1; Kimi K2.6: input dominates only above 4:1; Qwen3-Coder Next: input dominates only above 7:1
5. Latency			
PASS	A-04	Section 5.2's "∼5 μs/km in fibre"	4.89505
PASS	A-05	The latency-spectrum figures in section 5.3 are mutually consistent and correctly ordered	ordering holds; total span 3.33e+07x (7.5 orders of magnitude)
INFO	A-06	The report's own system sits at the far end of the latency spectrum	600 s vs 3 ns = 2.0e+11x
4. Risk			
PASS	A-07	McLean & Pontiff (2016) arithmetic quoted in section 4.7: a 58% post-publication decline minus a 26% out-of-sample decline gives the 32% attributed to publication-informed trading	32
PASS	A-08	Hou, Xue & Zhang (2020) failure rates quoted in section 4.7	294 of 452 anomalies fail the single test; 371 of 452 fail the multiple-testing hurdle (77 additional casualties)
PASS	A-09	Section 4.4's market-wide circuit-breaker (Rule 80B) thresholds are internally consistent	Level 1: 7% – 15-minute halt if before 15:25 ET; none at/after; Level 2: 13% – 15-minute halt if before 15:25 ET; none at/after; Level 3: 20% – halt for the remainder of the day, at any time

Status	ID	Claim	Measured
3.6 Regulatory			
PASS	`A-10`	Section 3.6's pattern-day-trader test is a CONJUNCTION of two conditions, and the report states both	4 day trades in 50 total = 8% -> triggers; 4 in 200 total = 2% -> does not trigger

13 Appendix B — Detail for every finding and caveat

Full method, expected value, measured value and recommendation for each of the three errors and nine caveats.

V-08b — 2.2 Volatility estimation

Claim under test. Range estimators are biased LOW on real bars: a bar built from finitely many prints under-observes the true high and low
Method. MC sweep over intraday observations per bar; slope compared against the Broadie-Glasserman-Kou continuity correction
Expected. theory: bias ~ -1.46*m^(-1/2) **Measured.** 100 obs: -12.0%; 200 obs: -9.1%; 400 obs: -6.0%; 800 obs: -4.5%; 1600 obs: -3.0%; 3200 obs: -2.2%; 6400 obs: -1.7%; fitted slope -1.21
The report lists Parkinson and Garman-Klass without this caveat. The unbiasedness proofs assume a continuously observed path; a real bar is a finite sample of prints. The fitted $m^{-1/2}$ slope (-1.21) is within ~15% of the Broadie-Glasserman-Kou prediction (-1.46), confirming the mechanism. Operationally: a thinly traded ETF whose bar holds ~100 prints yields a Parkinson variance about 12% too low. Section 2.3 divides by this number, so the error becomes systematic over-leverage. Use liquid instruments, or calibrate a discretisation correction from observed prints per bar.

V-12 — 2.2 Volatility estimation

Claim under test. Naive $\sqrt{252}$ (or $\sqrt{12}$) annualisation overstates the Sharpe ratio when returns are positively autocorrelated
Method. Lo (2002) correction factor at several AR(1) coefficients, plus MC confirmation
Expected. overstatement grows with rho **Measured.** rho=-0.2: -16.9%; rho=+0.0: -0.0%; rho=+0.1: +9.6%; rho=+0.2: +20.3%; rho=+0.3: +32.5%
The report says annualisation 'assumes i.i.d.; autocorrelation biases it' but gives no magnitude. At $\rho = 0.2$ – unremarkable for a daily trend strategy – naive scaling inflates the Sharpe by 20% (MC-confirmed). Any strategy whose Sharpe clears a hurdle only after naive annualisation should be re-tested with the Lo correction. Smoothed or illiquid marks push ρ higher still.

V-14 — 2.11 Backtest math

Claim under test. Applied to discrete simple returns, $g \approx \mu - \frac{1}{2}\sigma^2$ is a truncated expansion whose error is material at high volatility
Method. MC across sigma from 5% to 80% annual, 2M lognormal draws each
Expected. error -> 0 as sigma -> 0 **Measured.** sigma=5%: err -0.0001; sigma=10%: err -0.0004; sigma=20%: err -0.0019; sigma=40%: err -0.0127; sigma=60%: err -0.0440; sigma=80%: err -0.1078
The report gives $g \approx \mu - \frac{1}{2}\sigma^2$ without saying whether μ, σ are Ito parameters (where it is exact, V-13) or sample moments of simple returns (where it is not). At the ~20% volatility of a typical ETF sleeve the error is 0.19 pp – small but not negligible against the 3.78 pp that V-15 attributes to vol targeting. At 80% volatility it reaches 10.8 pp, or 82% of the geometric return itself. Use the formula as intuition for why vol targeting aids compounding; never as the P&L accounting identity. The backtester must compound realised returns.

K-02 — 2.3 Volatility targeting

Claim under test. A leverage cap $\sum |w| \leq L$ makes the vol target an upper bound, not an equality
Method. same simulation with cap L=1.0
Expected. target 15.0% **Measured.** realised 14.33% (cap binds 22% of the time)
The report presents the vol target and the leverage cap as independent constraints. They are not: whenever the cap binds, realised volatility falls below target (here 14.33% vs 15.0%). In calm regimes – exactly when σ^*/σ is largest – the cap silently converts the strategy from vol-targeted to constant-leverage. The system should log cap-binding frequency as a first-class monitoring metric, because a strategy that spends most of its life against the cap is not the strategy that was backtested.

P-13 — 2.4 Portfolio construction

Claim under test. Quadratic transaction costs do NOT create a no-trade region; proportional (L1) costs do
Method. sweep the starting weight away from the frictionless optimum and measure the traded amount under each cost model
Expected. report claims quadratic costs give a no-trade region **Measured.** quadratic: trade > 0 at every nonzero drift (min traded 3.98e-03); L1: exactly zero trade until drift reaches 0.04
Correction required. Section 2.4 states that quadratic costs 'give a closed form AND a no-trade region'. The closed form is correct (P-12); the no-trade region is not. A smooth quadratic penalty has zero derivative at zero trade, so the first-order condition always prescribes a strictly positive trade – the solution shrinks toward w_0 (partial adjustment) but never stops. A no-trade region requires a cost function with a KINK at zero, i.e. proportional/L1 costs, whose subgradient interval $[-c, c]$ absorbs small deviations. This is not pedantry: it changes the implementation. If the system is built on the quadratic model expecting bands to appear, it will rebalance every single day and bleed the cost the band was meant to save. Real spreads and commissions ARE proportional, so the L1 model is also the more faithful one. Recommended fix: either state the objective with an L1 term, or keep the quadratic form and impose the no-trade band as an explicit separate rule.

Q-02 — 2.7 Risk metrics

Claim under test. The report's $\mu + z_\alpha \sigma$ is ambiguous without a sign convention
Method. evaluate both readings at the same parameters
Expected. one unambiguous number **Measured.** return-quantile reading -0.0274; loss-quantile reading +0.0284
Section 2.7 gives parametric VaR as $\mu + z_\alpha \sigma$ without stating whether α is the tail probability or the confidence level, or whether VaR is reported as a positive loss or a negative return. The two readings differ by roughly 0.0558 here – a sign flip on the risk number itself. This is not a theoretical quibble: a sign error in a risk limit is precisely the class of defect the report's section 4.3 pre-trade controls exist to prevent, and it must be pinned down by a property-based test (section 6.1) asserting VaR > 0 and VaR $\leq$ CVaR under the house convention.

Q-08 — 2.7 Risk metrics

Claim under test. Cornish-Fisher is not monotone in the quantile for large skewness, so it can report a SMALLER loss at a HIGHER confidence level
Method. scan z in [0.5, 3.5] at several (skew, excess kurtosis) pairs and test for a decreasing segment
Expected. quantile function must be non-decreasing in z **Measured.** non-monotone at (skew, excess kurt) = [(-3.0, 12.0)]
The report lists Cornish-Fisher without its domain of validity. It is an asymptotic expansion, valid for mild non-normality only; at the skewness levels typical of an option-overlay or short-vol sleeve (section 4.6's XIV example) it can invert, reporting a smaller loss at 99% than at 95%. The runtime must assert monotonicity of the quantile function – a natural Hypothesis property test (section 6.1) – and fall back to historical or filtered-historical simulation when the assertion trips.

S-04 — 2.8 Statistical validation

Claim under test. γ_4 in the PSR formula must be NON-EXCESS kurtosis, and the size of the error depends on the return frequency
Method. recompute the Sharpe standard error under both kurtosis conventions across sampling frequencies, at a fixed annualised Sharpe of 0.8
Expected. identical results under either convention **Measured.** daily: -0.1%; weekly: -0.5%; monthly: -1.9%; quarterly: -5.1%
Section 2.8 defines $\hat{\gamma}_4$ only as 'kurtosis'. Both conventions are in common use and `scipy.stats.kurtosis` returns the EXCESS value by default, so the natural implementation is the wrong one. The error scales with the SQUARE of the per-period Sharpe, which makes it nearly invisible on daily data (-0.1%) but material on monthly data (-1.9%) and worse on quarterly (-5.1%). Since fund-style track records are almost always evaluated monthly, the bug would surface precisely when the stakes are highest – and it

understates the standard error, which inflates PSR and DSR and lets strategies through the promotion gate that should have been rejected. The direction of the error is the dangerous one. Separately, note that the SKEWNESS term dominates at daily frequency: a realistic skew of -0.8 changes the standard error by $+2.0\%$ on daily data, an order of magnitude more than the kurtosis term. Fix: define $\gamma_4 = E[(r-\mu)^4]/\sigma^4$ explicitly in the report, and unit-test that a Gaussian sample gives $\gamma_4 \approx 3$ and that the denominator reduces to $\sqrt{1+\widehat{SR}^2/2}$.

S-14 — 2.8 Statistical validation

Claim under test. A PBO value computed from a single backtest matrix has very large sampling error and cannot be read as a precise number
Method. 24 independent null datasets at each (T, N); report the spread of the resulting PBO estimates
Expected. PBO should concentrate near 0.5 under the null **Measured.** T=1000, N=40: mean 0.49, sd 0.19; T=2500, N=40: mean 0.52, sd 0.20; T=1000, N=200: mean 0.47, sd 0.16; T=5000, N=200: mean 0.52, sd 0.15
The report presents PBO as a scalar to be compared against a threshold. In practice, at the data sizes a solo operator has, the estimator's own standard deviation under the null is around 0.19 at T=1000, N=40. Two honest runs of the same worthless strategy family can therefore return PBO = 0.25 and PBO = 0.75. Consequences for the build: (i) do not gate promotion on a single PBO point estimate; (ii) report a bootstrap confidence interval alongside it; (iii) prefer longer histories and larger strategy families, since the spread falls to 0.15 at T=5000, N=200. This does not weaken the case for PBO – it is still the right diagnostic – but a threshold rule written against a noisy statistic is a false gate, and false gates are worse than no gate because they are trusted.

E-03 — 2.9 Transaction cost & market impact

Claim under test. $\kappa \approx \sqrt{\lambda\sigma^2/\eta}$ is the continuum limit of the exact κ, and drops the $\tilde{\eta} = \eta - \gamma\tau/2$ correction
Method. compare the report's kappa against the exact root of cosh(kappa tau) = 1 + kappa_tilde^2 tau^2/2, across N; also report the resulting objective-function penalty
Expected. exact kappa **Measured.** N=5: kappa error -0.49%, cost penalty +9.76e-09; N=10: kappa error -0.25%, cost penalty +2.57e-09; N=25: kappa error -0.10%, cost penalty +4.18e-10; N=50: kappa error -0.05%, cost penalty +1.05e-10; N=200: kappa error -0.01%, cost penalty +6.56e-12; N=1000: kappa error -0.00%, cost penalty +2.62e-13
The approximation is sound but its conditions should be stated. Two distinct corrections are dropped. First, the exact κ solves a transcendental equation whose solution approaches $\tilde{\kappa}$ only as $\tau \to 0$. Second, the denominator should be $\tilde{\eta} = \eta - \gamma\tau/2$, not η: the permanent-impact parameter partially offsets the temporary one because half of each trade's permanent impact is paid by the trader's own remaining inventory. At a fine grid (N=1000) the error in κ is -0.00% and irrelevant; on a coarse schedule (N=5, i.e. five slices – a realistic daily-ETF execution) it is -0.49%. Crucially, the OBJECTIVE penalty is tiny in every case (9.8e-09 at worst), because the cost surface is very flat near its optimum. The practical conclusion for the report is reassuring and worth stating explicitly: execution-schedule optimisation is forgiving, so effort spent calibrating impact parameters precisely is better spent elsewhere – which reinforces the report's own section 5.7 argument against premature optimisation.

M-04 — 6.5 Pairs / statistical arbitrage

Claim under test. The report's half-life $= -\ln 2/\lambda$ systematically OVERSTATES the half-life, and the error grows with the speed of mean reversion
Method. same simulation; relative error of the report's formula against the known truth
Expected. half-life estimate should equal ln(2)/theta **Measured.** theta=0.02 (true HL 34.7): report 34.7 (-0%); theta=0.05 (true HL 13.9): report 14.2 (+2%); theta=0.10 (true HL 6.9): report 7.3 (+6%); theta=0.25 (true HL 2.8): report 3.1 (+14%); theta=0.50 (true HL 1.4): report 1.8 (+27%); theta=1.00 (true HL 0.7): report 1.1 (+58%)
Correction required. Section 6.5 states 'Estimate via OLS of $\Delta X_t = \lambda X_{t-1} + c + \varepsilon_t$; then $\theta = -\lambda$ and half-life $= -\ln 2/\lambda$'. The exact discretisation of the OU process gives $\lambda = e^{-\theta\Delta t} - 1$, so the correct inversion is $\theta = -\ln(1+\lambda)/\Delta t$ and half-life $= -\Delta t \ln 2/\ln(1+\lambda)$. The report's form is the first-order

Taylor expansion, accurate only when $\theta\Delta t \ll 1$. For slow reversion (theta ≤ 0.05) the error is under 2.3% and harmless. For the fast reversion that actually makes a pairs trade worth doing (theta ≥ 0.25) it reaches 58%. The error is one-directional: the estimate is always too LONG. Operationally that means holding period limits, time-based stops and capital-allocation horizons are all set too generously, on the strategies where the mis-estimate is worst. The fix is one line of code and removes the bias entirely.

A-01 — 1. Alternative and cheaper models

Claim under test. Section 6.7's claim of a "∼90–98%-cheaper cache rate on DeepSeek/GLM/Kimi"
Method. compute 1 - (cache-hit price / cache-miss price) from the report's own section 1.2 pricing table
Expected. 90-98% discount across DeepSeek, GLM and Kimi **Measured.** DeepSeek V4 Flash 98%; DeepSeek V4 Pro 99%; Kimi K2.6 / K2.7 Code 80%; Kimi K2.5 75%; Kimi K3 90%; GLM-4.6 82%; GLM-4.5-Air 85%
Internal inconsistency. The 90–98% range holds only for DeepSeek (98% and 99%). For the GLM and Kimi families the same table gives discounts of 75%–90% – real and worth having, but not the order of magnitude claimed. The two statements are computed from the same table and contradict each other. This matters because section 6.7 presents prompt caching as the primary cost-control lever: if the actual saving on a GLM coding plan is 82% rather than 98%, the residual spend is nine times larger than the sentence implies. Recommended fix: state "up to ∼98% on DeepSeek, ∼75%–90% on GLM/Kimi".

14 Appendix C — Verification suite structure

Module	Report sections	Checks
`v01_returns_vol.py`	2.1, 2.2, 2.11	19
`v02_sizing.py`	2.3, 2.6	11
`v03_portfolio.py`	2.4	13
`v04_risk.py`	2.7	9
`v05_validation.py`	2.8	15
`v06_execution.py`	2.9, 2.10	12
`v07_meanrev.py`	6.5	7
`v08_ml_regime.py`	2.5, 2.12	8
`v09_report_arithmetic.py`	1, 3.6, 4, 5	10

Verification methods used, in order of strength:

1. **Symbolic proof** (`sympy`) — the strongest available. Used for the Markowitz closed form, both Kelly forms, the fractional-Kelly growth identity, the Almgren–Chriss stationarity condition, Kyle's lambda, and the square-root law. A symbolic result is an identity, not a coincidence.
2. **Independent numerical solve** — re-derive the answer with an optimiser that assumes nothing about the closed form, then compare. Used for Almgren–Chriss (agreement to 6.6×10^{-13}), ERC, the transaction-cost objective, and the Minimum Track Record Length.
3. **Monte Carlo against a known generating process** — construct data whose true parameters are known by design, then check the estimator recovers them. Used for every volatility estimator, the Sharpe standard error, the False Strategy Theorem, PBO calibration, Roll's spread, and the OU half-life.
4. **Null-calibration testing** — feed a procedure data containing no signal and confirm it says so. This is how M-05 (57% versus a nominal 5%) and S-15 (purged CV) were found, and it is the most productive single technique in the suite.
5. **Richardson extrapolation** — where an estimator is only unbiased in a limit, sweep the discretisation and extrapolate. Used to separate genuine estimator bias from sampling artefact in the range volatility estimators.

Determinism

Master seed `20260811`. Every random stream in the suite is derived from it via `numpy.random.default_rng([MASTER_SEED, stream_id])`, so streams are independent and reproducible. Re-running the suite reproduces every number in this document exactly. No verification figure in the prose is hardcoded — all tables and counts are generated from the suite's JSON output.